

Abstract:

This project presents the design and development of an online pharmacy platform that enables users to conveniently access medicines and healthcare products through a digital interface. The platform allows users to browse available medications, search using filters, place orders, and manage their purchases efficiently, while administrators can manage products, inventory, and orders.

The system integrates an AI-powered assistant using Google Gemini API to provide real-time responses to user inquiries and deliver general information about medicines, including usage instructions and basic details. The assistant focuses on answering user questions and providing reliable information.

With a focus on usability, performance, and scalability, the platform is designed to handle multiple users simultaneously while maintaining a responsive and user-friendly interface. The system aims to improve accessibility to pharmaceutical services and provide a reliable and efficient solution for online healthcare access.

DEDICATION:

This project is dedicated to all individuals seeking accessible and reliable healthcare solutions through digital platforms. It is also dedicated to those who aim to improve the quality of life by leveraging technology in the pharmaceutical field.

We would like to express our sincere gratitude to **Dr. Safaa Magdy** for her continuous support, guidance, and valuable feedback throughout the development of this project.

Table of Contents :

Cover	0
Abstract	1
Dedication	2
Chapter 1 (Introduction)	6-11
1.1 Problem Statement	7
1.2 Objectives	8
1.3 Scope of Work	9
1.4 Overview of Document	10
Chapter 2 (Background & Literature Review)	11-16
2.1 General Constraints	12
2.1.1 Time	12
2.1.2 Data Collection	12
2.2 Project Description	13
2.3 Assumptions and Constraints	13-14
2.4 Background and Context	14
2.5 Project Benefits	15-16

Chapter 3 (Methodology)	17-50
3.1 Software Tools	18
3.2 Software Requirements	19
3.2.1 Functional Requirements	19-27
3.2.2 Non-Functional Requirements	28
3.3 Software Design	29-50
3.3.1 Class Diagram	29
3.3.2 Use Case Diagram and Scenarios	30-37
3.3.3 Sequence Diagram	38-42
3.3.4 Activity Diagram	43-50
Chapter 4 (Implementation)	51-58
4.1 Software Architecture	52-54
4.2 Database Architecture.....	55-58
Chapter 5 (Results and Discussion)	59-66
5.1 Results	60-65
5.1.1 Expected Results	60-62
5.1.2 Actual Results	63-65
5.2 Discussion	66
Chapter 6 (User Interface)	67-77

Chapter 7 (Conclusion)	78-80
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CHAPTER 1

Introduction

1.1 Problem Statement:

In the rapidly growing digital healthcare sector, many patients and customers face difficulties in accessing pharmaceutical products in a convenient and timely manner. Traditional pharmacies require physical visits, which can be challenging for elderly people, patients with chronic illnesses, or individuals living in remote areas.

Pharmacies, on the other hand, often struggle with efficient inventory management, order tracking, and customer communication when relying on manual or semi-digital systems.

Customers frequently encounter challenges such as:

- 1) Difficulty in finding required medicines quickly.
- 2) Limited access to reliable pharmacy services outside working hours.
- 3) Lack of clear information about medicine availability and pricing.

At the same time, pharmacies face challenges including:

- 1) Inefficient management of medicine inventory and orders.
- 2) Limited digital presence and customer reach.
- 3) Difficulty in maintaining organized records of customers and transactions.

This project aims to address these issues by developing a comprehensive and scalable online pharmacy web application that connects customers with pharmacies through a secure and user-friendly platform. The system is designed to improve accessibility, efficiency, and reliability for both customers and pharmacy administrators.

1.2 Objectives:

The primary objective of this project is to design and develop a comprehensive web-based Online Pharmacy system that enables users to access pharmaceutical products easily and efficiently through the internet. The system aims to provide a reliable digital solution that improves the interaction between customers and pharmacies while reducing the dependency on traditional, in-person pharmacy visits.

This project also seeks to enhance the overall pharmacy service experience by streamlining operations such as medicine management, order processing, and user account handling. By leveraging modern web technologies, the system ensures better organization, faster service delivery, and improved accuracy in handling pharmaceutical data.

The specific objectives of this project include:

- * To design a simple and intuitive user interface that allows customers to browse and purchase medicines with minimal effort.
- * To develop a centralized system that enables pharmacies to manage medicine inventory and customer orders efficiently.
- * To reduce delays and inconvenience associated with traditional pharmacy services.
- * To ensure secure handling of user data through proper authentication and access control mechanisms.
- * To provide a scalable system that can accommodate future growth and additional functionalities.

1.3 Scope of Work:

This project covers the design and development of a complete web-based Online Pharmacy system that aims to support the essential processes involved in online pharmaceutical services. The system provides a digital platform that enables organized interaction between customers and pharmacies while maintaining clarity in system operations and responsibilities.

The scope of work includes analyzing system requirements, designing the overall system structure, and implementing both frontend and backend components. The frontend focuses on delivering a clear and responsive user interface, while the backend is responsible for handling business logic, data storage, and system security.

The project also involves defining different user roles within the system and ensuring that each role has appropriate access to system features. Special attention is given to data integrity, system reliability, and performance to ensure consistent operation under different usage conditions.

This project is limited to a web-based environment and focuses solely on the core functionality of an online pharmacy. Features outside the defined system boundaries, such as mobile application development or integration with external healthcare systems, are not included within the scope of this project.

1.4 – Overview of remainder of document:

- * **Module Descriptions:** Detailed explanation of each major system module in the online pharmacy system (e.g., user management, medicine management, order processing, and pharmacy administration), including the role of each module and the flow of data between them.
- * **Component Interactions:** Overview of how different system components interact with each other, such as the interaction between user accounts, medicine inventory, and order processing modules.
- * **User Interface Layout:** Descriptions of the main pages and screens of the system, including their content, layout structure, and user navigation flow.
- * **Functional Requirements:** A detailed list of all functional requirements expected from the system, describing the actions that users and administrators can perform within the online pharmacy platform.
- * **Non-Functional Requirements:** Definitions of system quality attributes such as performance, scalability, security, usability, and availability.
- * **Use Case Diagrams and Scenarios:** Visual and textual representations of user interactions with the system, illustrating how customers and administrators use the system to achieve specific goals.
- * **Special Requirements:** Any additional technical or operational constraints related to the system, such as data privacy, security regulations, and system compatibility considerations.

CHAPTER 2

Background & Literature review

2.1 General Constraints

Hardware Constraints

- Availability of a computer device or a smart phone to access the online pharmacy system.

Software Constraints

- Availability of a stable Wi-Fi connection or reliable mobile data.
Secure internet connection to ensure safe data transmission.

2.1.1 Time

- No time constraints, as users can access the online pharmacy system at any time.

2.1.2 Gathering Data

- A sufficient amount of data is required to operate the system efficiently.
- Medicine data, user data, and order data are required for full system functionality.

2.2 Project Description

This project is a web-based online pharmacy system designed to provide a digital platform that connects customers with pharmacies. The system aims to simplify the process of searching for medicines and placing orders by offering a secure, reliable, and user-friendly environment.

Pharmacies can manage their accounts, add and update medicine information, monitor inventory levels, and handle customer orders through a centralized system. The platform enables pharmacies to maintain accurate records and ensure medicine availability.

Customers, on the other hand, can browse available medicines, view detailed product information, add items to their cart, place orders, and track their order status. The system is designed to support a smooth and efficient purchasing experience.

Additionally, an administrative interface is provided to monitor system activity and ensure data integrity. The system is developed with scalability, security, and usability in mind and follows modern web development standards.

2.3 Assumptions and Constraints

Assumptions

- * Users have access to stable internet connectivity and modern web browsers.
- * Users possess basic digital skills to navigate the system and perform common tasks such as registration, login, and ordering.

- * Users provide accurate and valid information during registration and transactions.
- * Mobile users access the system through responsive web interfaces.

Constraints

- * The system operates strictly within a web-based environment.
- * Development must be completed within the defined project timeline.
- * Sensitive user data (e.g., passwords) must be securely encrypted.
- * System functionality depends on server availability and internet connectivity.
- * Language support is limited to English in the initial release.

2.4 Background and Context

With the increasing shift toward digital healthcare services, many individuals face challenges in accessing medicines conveniently and efficiently. Traditional pharmacy services often require physical visits, which may not be suitable for all users.

This project was initiated to address these challenges by developing a web-based online pharmacy system that provides essential pharmaceutical services through a centralized platform. The system supports medicine management, order processing, and user interaction in a unified digital environment.

The online pharmacy system aims to improve accessibility, enhance user experience, and support the digital transformation of healthcare services.

2.5 Project Benefits

The proposed system offers the following benefits:

For Pharmacies:

- 1) Easy management of medicines and inventory through a centralized system.
- 2) Efficient handling of customer orders.
- 3) Improved organization and record keeping.

For Customers:

- 1) Easy access to medicines through online browsing.
- 2) Convenient ordering and order tracking.
- 3) Reduced need for physical pharmacy visits.

For System Administrators

- 1) Clear visibility of system usage and activity.
- 2) Tools to monitor system performance and data integrity.

General Benefits

- * Improved accessibility to pharmaceutical services.
- * Enhanced efficiency and reliability of pharmacy operations.
- * Supports digital transformation in the healthcare sector.

CHAPTER 3

METHODOLOGY

3.1 Software Tools:

- * **Frontend:** React.js (JavaScript)
- * **Backend:** Django (Python)
- * **Database:** SQLite Database
- * **Machine Learning Model:** Django (Python)
- * **Third Party Integration:** Gemini API

3.2 Requirements:

3.2.1 Functional Requirements:

1) Admin

Function Name	Admin Login
Description	Admin logs into the platform using valid credentials to access system features.
Pre-Conditions	Admin must have a valid account.
Priority	High

Function Name	Admin Logout
Description	Admin can log out from their account safely.
Pre-Conditions	Admin must be logged in.
Priority	Medium

Function Name	Manage Categories
Description	Admin can add, edit, or delete medicine categories to organize products efficiently.
Pre-Conditions	Admin must be logged in.
Priority	High

Function Name	Manage Products
Description	Admin can add new products, update existing products, or delete products. Includes product details such as name, description, price, quantity, image, and expiry date.
Pre-Conditions	Admin must be logged in.
Priority	High

Function Name	Manage Users
Description	Admin can view, edit, or delete user accounts within the system. This helps control system access and maintain accurate user information.
Pre-Conditions	Admin must be logged in.
Priority	High

Function Name	Manage Orders.
Description	Admin can view all customer orders and update the order status by approving or rejecting orders. This helps ensure proper order tracking and controlled order processing.
Pre-Conditions	Admin must be logged in.
Priority	High

Function Name	Edit Profile
Description	Admin can update personal profile information such as name, email, and password. This helps keep account information up to date and secure.
Pre-Conditions	Admin must be logged in.
Priority	Medium

2) Customer

Function Name	Customer Register
Description	Allows a new customer to create an account by providing personal details. The system stores the information for future login.
Pre-Conditions	Customer must not have an existing account.
Priority	High

Function Name	Customer Login.
Description	Allows the customer to log in using email or phone number and password. Ensures secure access to customer features.
Pre-Conditions	Customer must have a registered account and enter valid information.
Priority	High

Function Name	Search Medicines.
Description	Allows customers to search for medicines by name, keyword, or category.
Pre-Conditions	Medicines must be available in the system.
Priority	Medium

Function Name	Add To Cart.
Description	Allows customers to add selected medicines to the shopping cart.
Pre-Conditions	Customer must be logged in.
Priority	High

Function Name	Make an Order.
Description	Allows customers to place an order for the selected medicines in the cart.
Pre-Conditions	Customer must be logged in and the cart must contain at least one medicine.
Priority	High

Function Name	Make Payment.
Description	Allows customers to complete the payment process for their orders using the available payment method.
Pre-Conditions	Customer must be logged in and an order must be created.
Priority	High

Function Name	Interact With Chatbot.
Description	Allows customers to interact with the chatbot to ask questions, get medicine information, and receive assistance during system usage.
Pre-Conditions	Customer must be logged in.
Priority	Medium

Function Name	Ask Doctor.
Description	Allows customers to send medical questions to a doctor for advice before purchasing medicines.
Pre-Conditions	Customer must be logged in.
Priority	Medium

-Edit Profile.

-Logout.

3) Store Manager

Function Name	Store Manager Login.
Description	Allows the store manager to sign in using email or username and password. Ensures secure access to store management features.
Pre-Conditions	Store manager must have a valid account.
Priority	High.

Function Name	Manage Stock Product.
Description	Allows the store manager to monitor medicine stock levels. The system generates alerts for low-stock products to prevent shortages.
Pre-Conditions	Store manager must be logged in.
Priority	High.

Function Name	Order Low Stock Product.
Description	Allows the store manager to place orders for medicines that are running low. Helps maintain sufficient stock levels and smooth store operations.
Pre-Conditions	Store manager must be logged in.
Priority	Medium.

-Edit Profile.

-Logout.

4) Doctor

-Login

Function Name	Answer Customer Questions.
Description	Allows the doctor to respond to customer medical questions submitted through the system. Provides medical guidance and advice before customers purchase medicines.
Pre-Conditions	Doctor must be logged in.
Priority	High.

-Edit Profile.

-Logout.

3.2.2- Non-Functional requirements:

1. Performance:

The system should respond to user actions (such as searching for medicines, adding items to cart, and placing orders) within 2 seconds under normal load conditions.

2. Availability:

The platform should maintain an uptime of 99.9% annually, minimizing downtime caused by maintenance or technical issues.

3. Security:

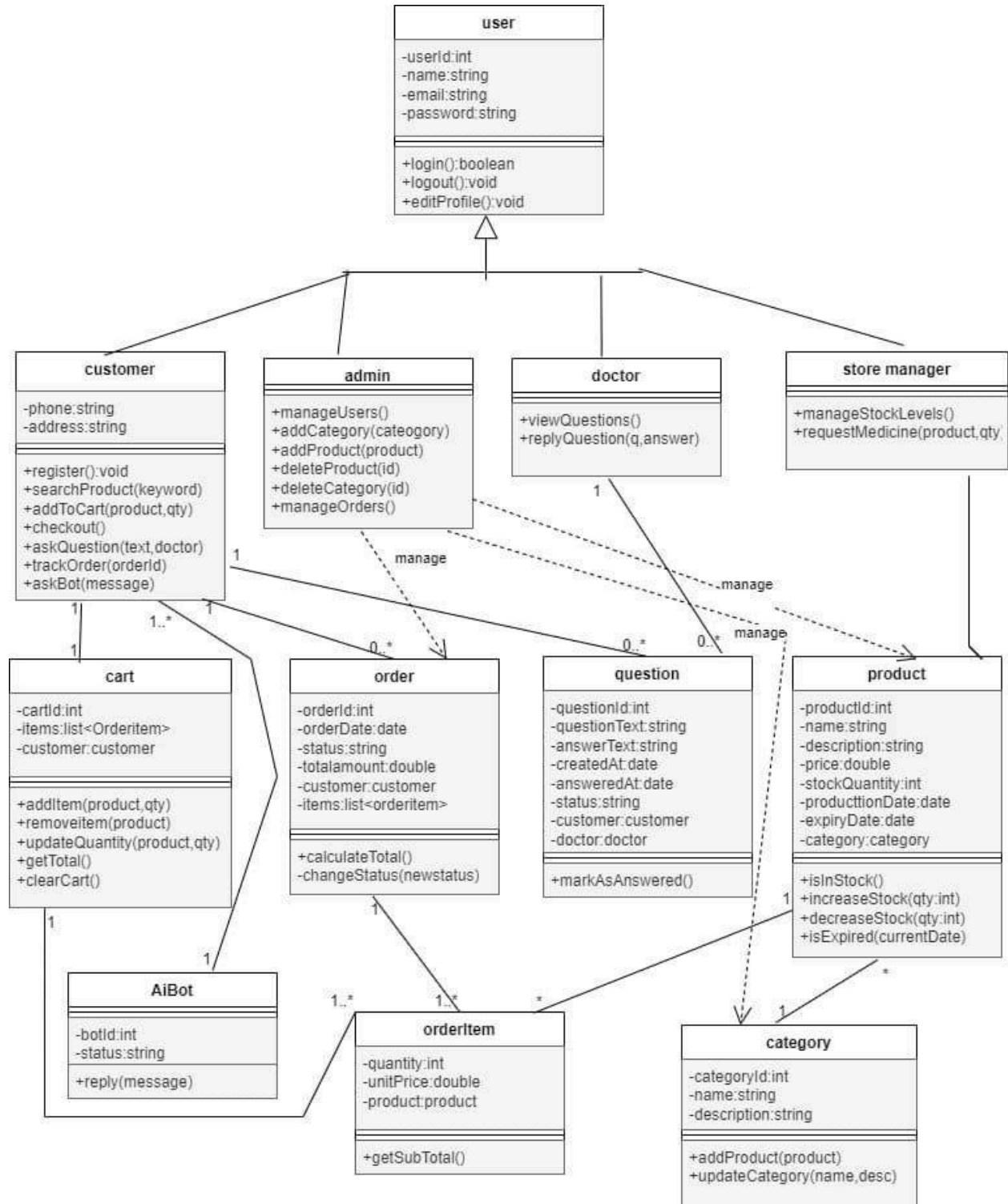
All sensitive data (such as user passwords and personal information) must be stored securely using encryption or hashed values to ensure data protection.

4. Usability:

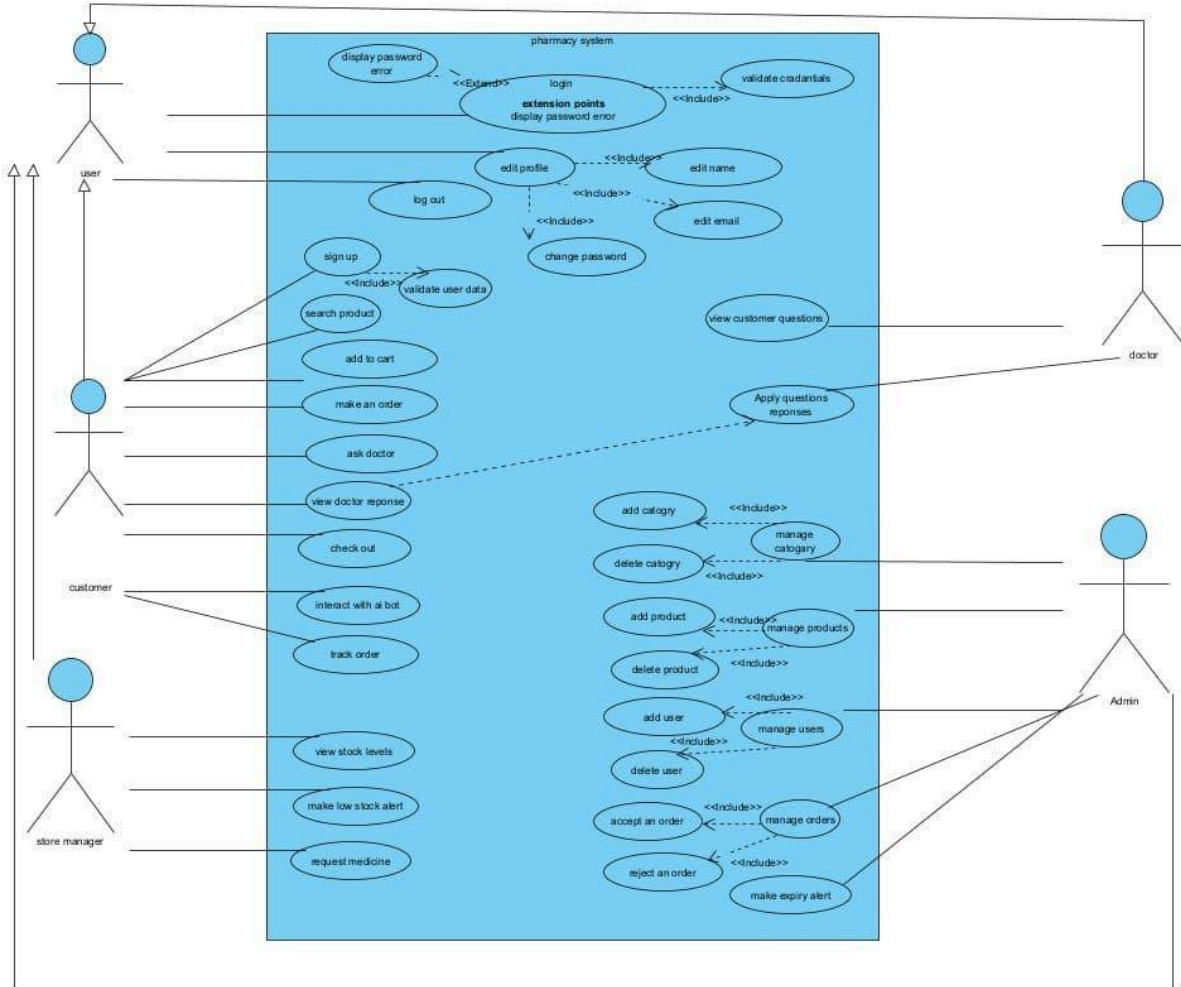
The system should have an intuitive and user-friendly interface, allowing users to complete common tasks (such as purchasing medicines or managing products) in less than three clicks.

3.3- Software Design:

3.3.1- Class Diagram:



3.3.2- Use Cases Diagram and Scenarios:



Goal Name	Sign Up
Goal	Allow users to create an account in the online pharmacy system.
Actors	Customer.
Pre-Conditions	User is not already registered in the system.
Postconditions	User account is successfully created.
Error Situations	*Missing required information *Invalid input data *Email or phone already exists
Standard Process	*User enters personal information. *System validates user data. *Account is created and confirmation is sent to the user.

Goal Name	Login
Goal	Authenticate users to access the system.
Actors	Customer – Admin – Store Manager – Doctor
Pre-Conditions	User has a valid account.
Postconditions	User is logged in successfully.
Error Situations	*Invalid credentials *Account not found
Standard Process	*User enters login credentials. *System verifies credentials. *User gains access to the system or receives an error message.

Goal Name	Logout
Goal	Allow users to securely exit the system.
Actors	Customer – Admin – Store Manager – Doctor
Pre-Conditions	User is logged in.
Postconditions	User session is terminated.
Error Situations	Session termination fails
Standard Process	*User enters personal information. *System validates user data. *Account is created and confirmation is sent to the user.

Goal Name	Search Product
Goal	Allow users to search for products in the online pharmacy system.
Actors	Customer.
Pre-Conditions	Products are available in the system.
Postconditions	Matching products are displayed to the user.
Error Situations	No products found.
Standard Process	*User enters product name, keyword, or category. *System searches the product database. *Matching products are displayed.

Goal Name	Manage Products
Goal	Allow the admin to manage products in the online pharmacy system.
Actors	Admin.
Pre-Conditions	Admin is logged in with valid credentials.
Postconditions	Product information is successfully added, updated, or deleted.
Error Situations	*Invalid product data *Missing required information *Product not found
Standard Process	*Admin selects the product management option. *Admin adds a new product or updates or deletes an existing product. *System validates the product information. *Product data is saved and the product list is updated.

Goal Name	Manage Users.
Goal	Allow the admin to manage Users in the online pharmacy system.
Actors	Admin.
Pre-Conditions	Admin is logged in with valid credentials.
Postconditions	User accounts are successfully viewed, updated, or deleted.
Error Situations	*User not found *Invalid user data *Unauthorized action
Standard Process	*Admin selects the user management option. *Admin views the list of users. *Admin updates user information or deletes user accounts when required. *System saves changes and updates user records.

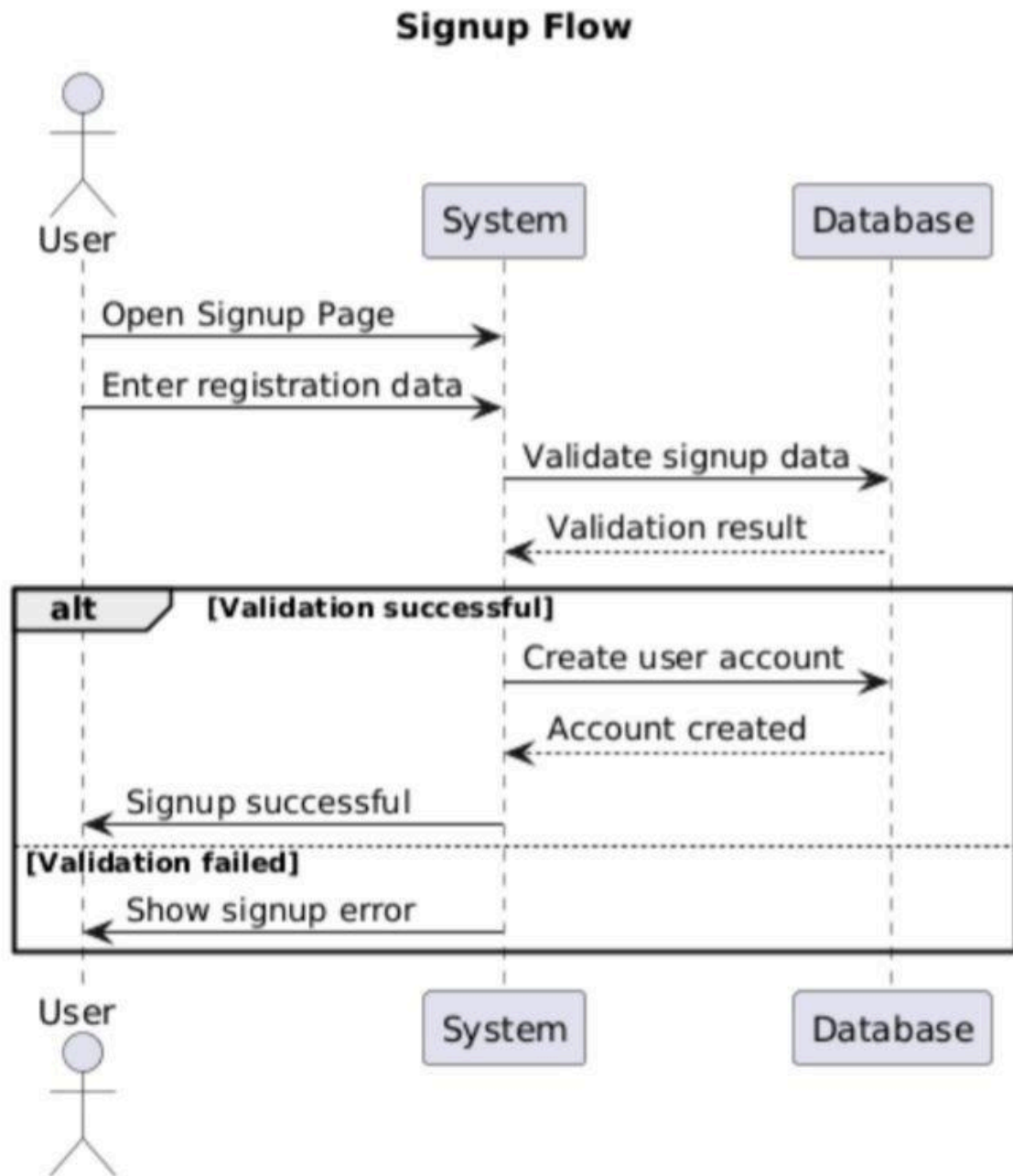
Goal Name	Manage Order.
Goal	Allow the customer to place an order for selected products in the online pharmacy system.
Actors	Customer.
Pre-Conditions	Customer is logged in and the shopping cart is not empty.
Postconditions	Order is successfully created and submitted for processing.
Error Situations	*Cart is empty *Product out of stock *Order creation failed
Standard Process	*Customer reviews the products in the shopping cart. *Customer confirms the order. *System checks product availability. *System creates the order and updates its status.

Goal Name	Make Low Stock Alarm.
Goal	Notify the store manager when product stock reaches a low level.
Actors	Store Manager.
Pre-Conditions	Store manager is logged in and products exist in the system.
Postconditions	Low stock notification is generated and displayed to the store manager.
Error Situations	*Product data not available *Notification generation failed
Standard Process	*System monitors product stock levels. *System detects products with low quantity. *System generates a low stock alert. *Store manager receives the notification.

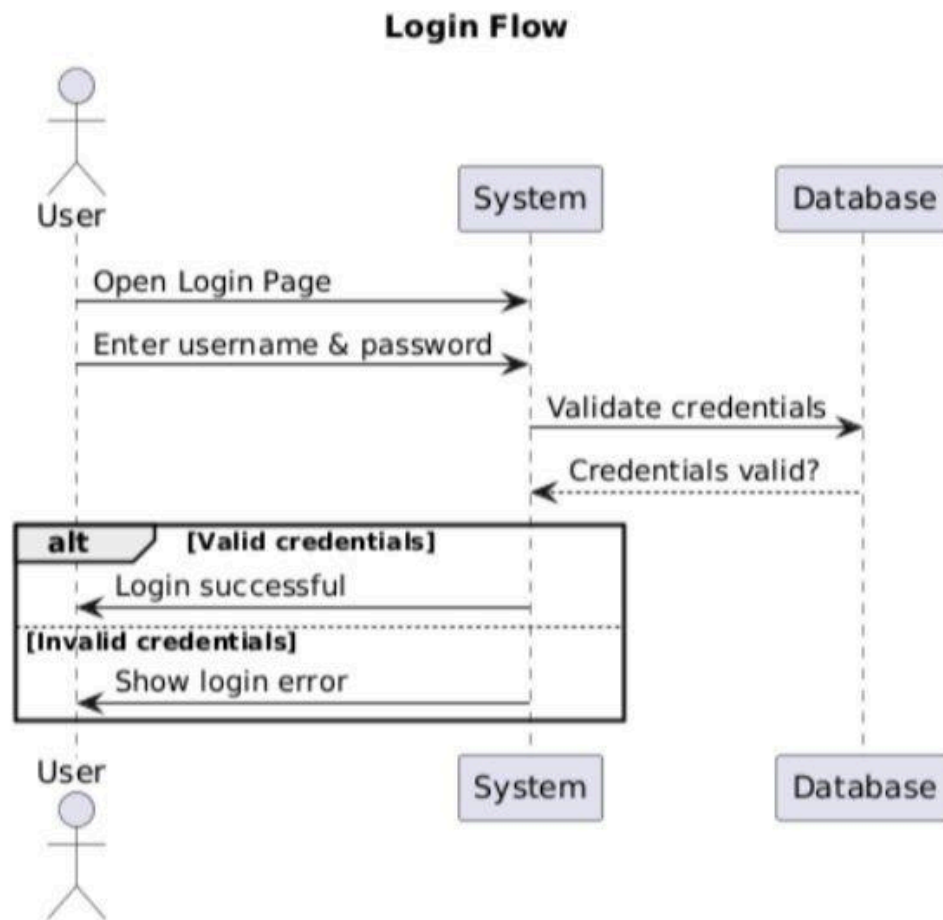
Goal Name	Apply Question Response.
Goal	Allow the doctor to respond to customer questions through the online pharmacy system.
Actors	Doctor.
Pre-Conditions	Doctor is logged in and customer questions are available.
Postconditions	Doctor response is successfully sent to the customer.
Error Situations	*Question not found *Response submission failed
Standard Process	*Doctor views the list of customer questions. *Doctor selects a question to respond to. *Doctor writes and submits the response. *System delivers the response to the customer.

3.3.3- Sequence Diagram:

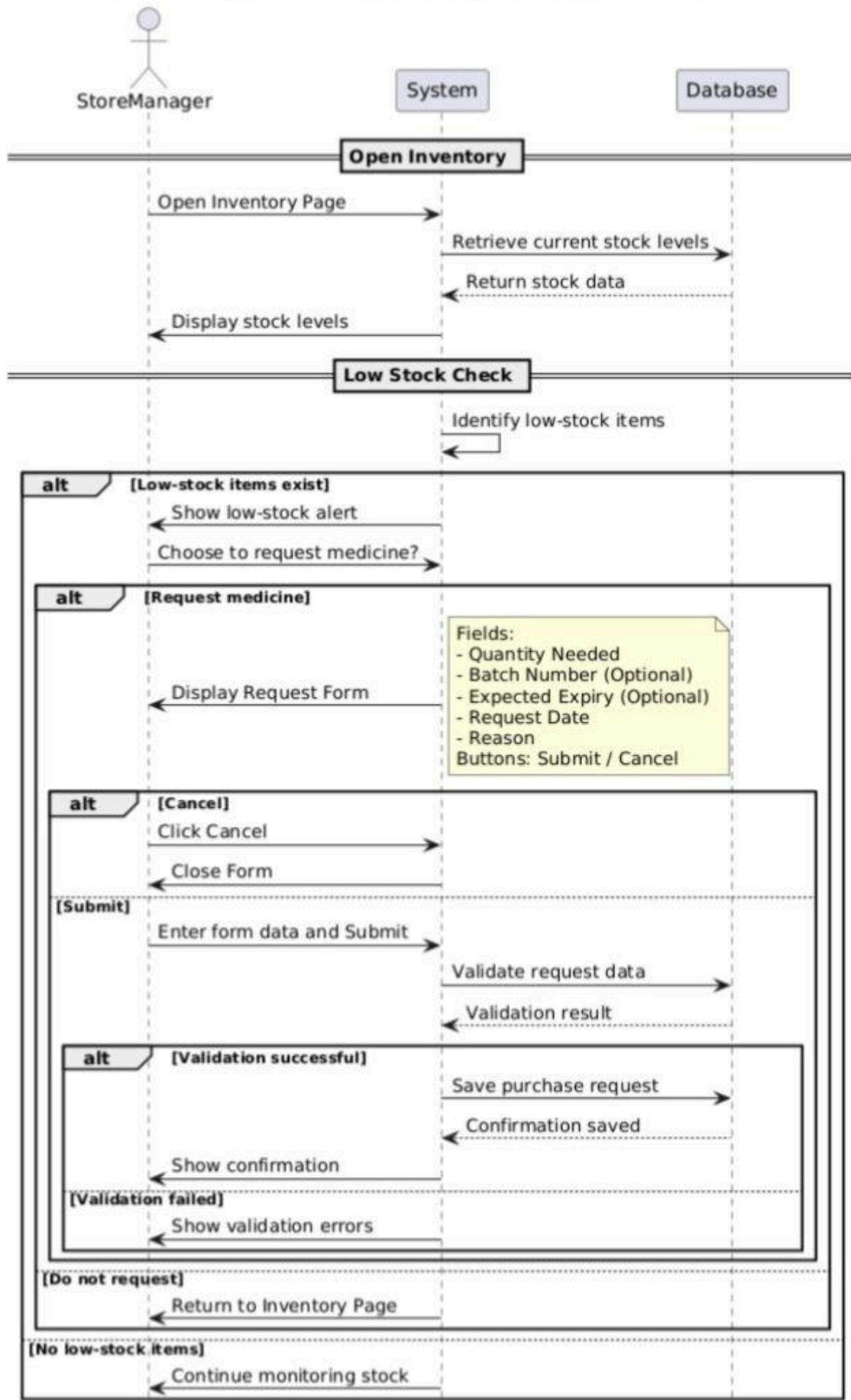
* sign up:



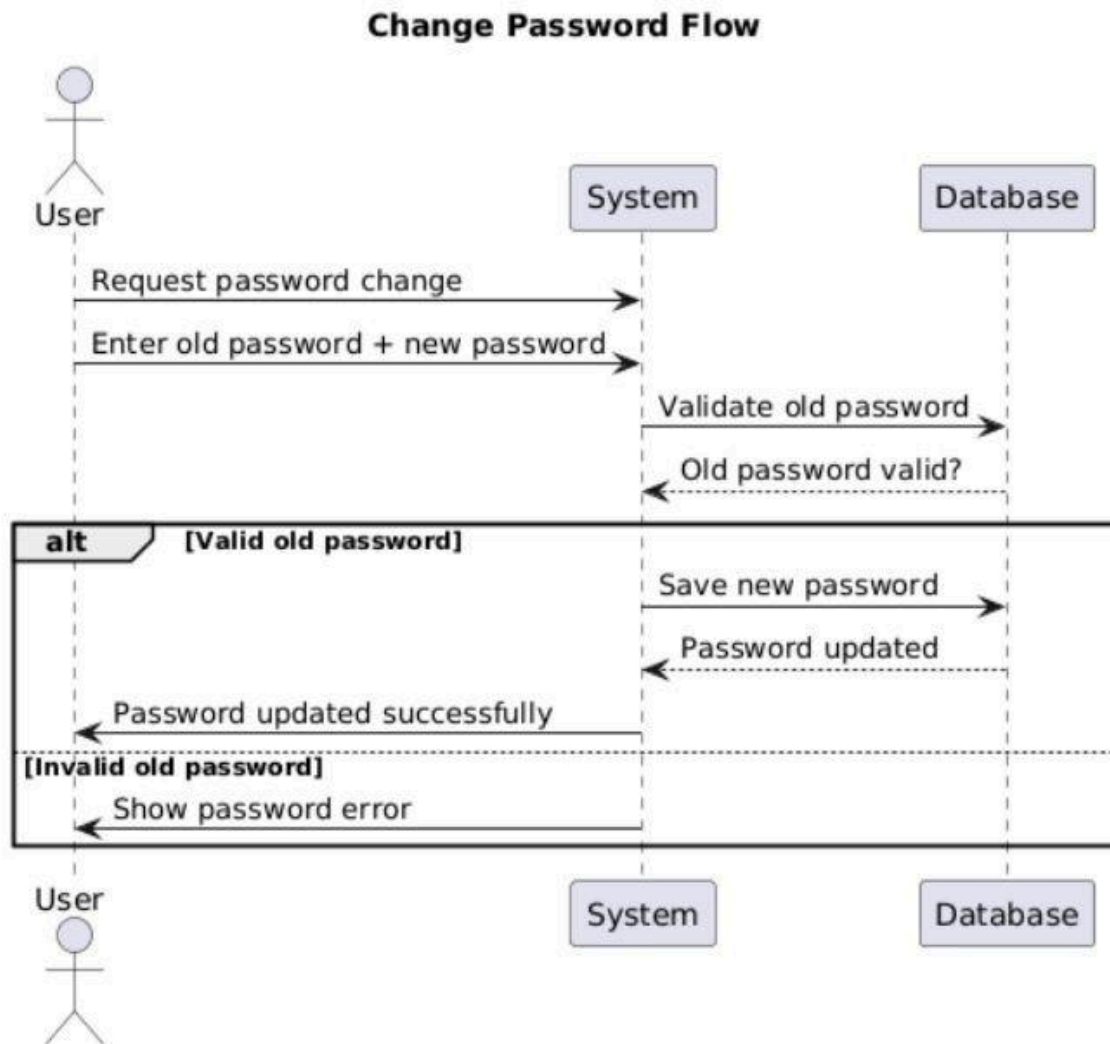
*** Login:**



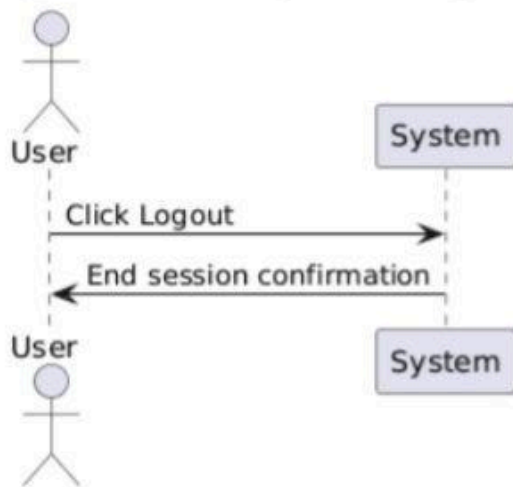
Store Manager - Inventory Management Sequence Diagram



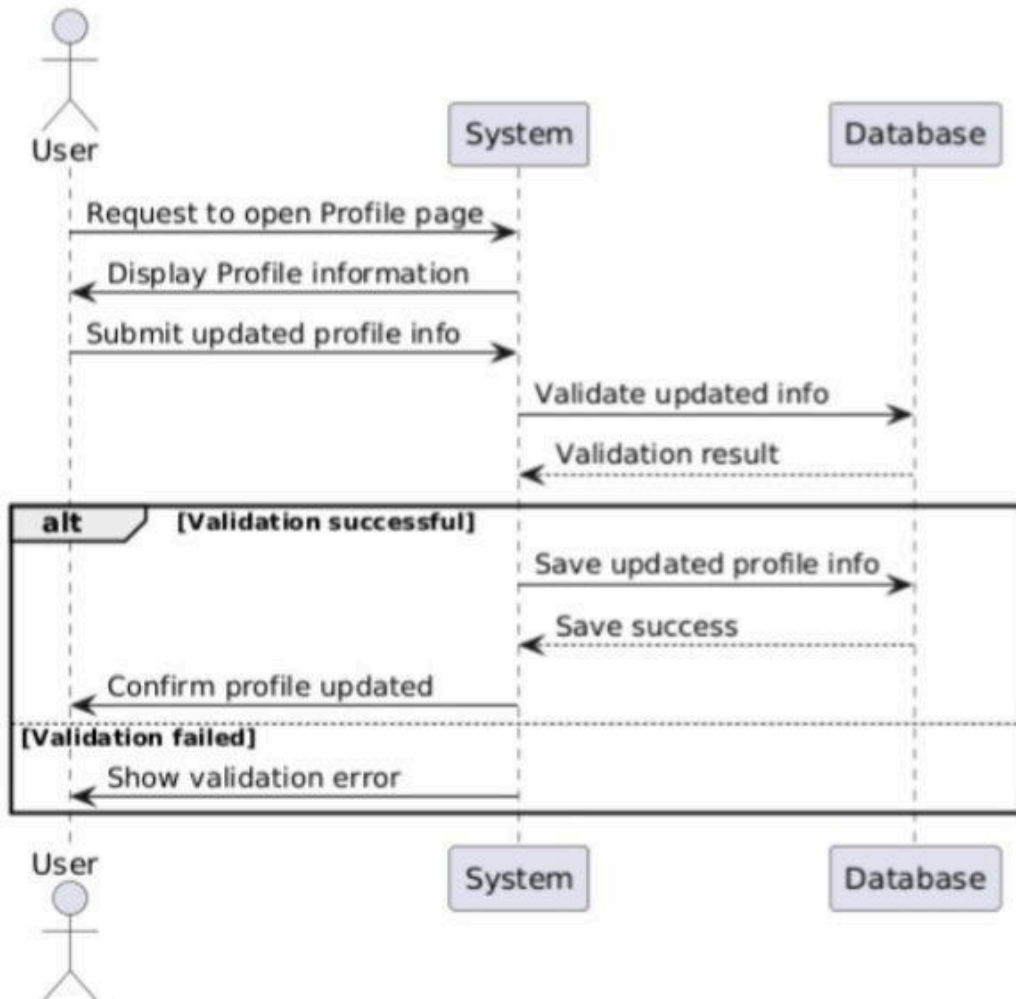
* Edit Profile & Logout:



Logout Flow - Sequence Diagram

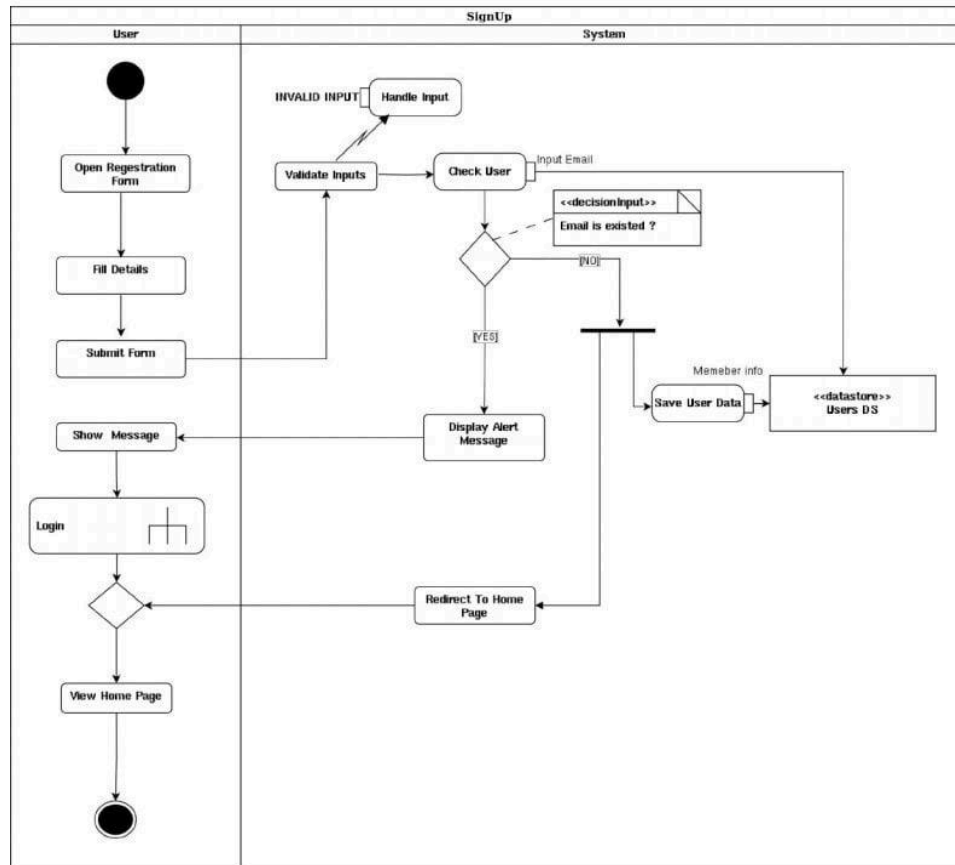


Edit Profile Flow

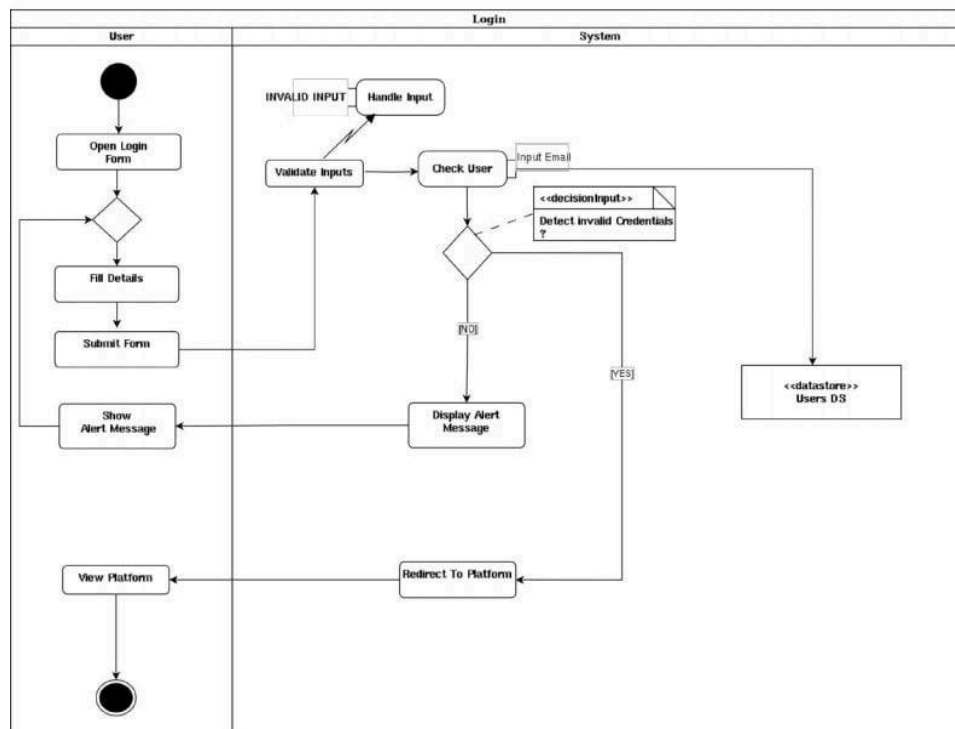


3.3.4- Activity Diagram:

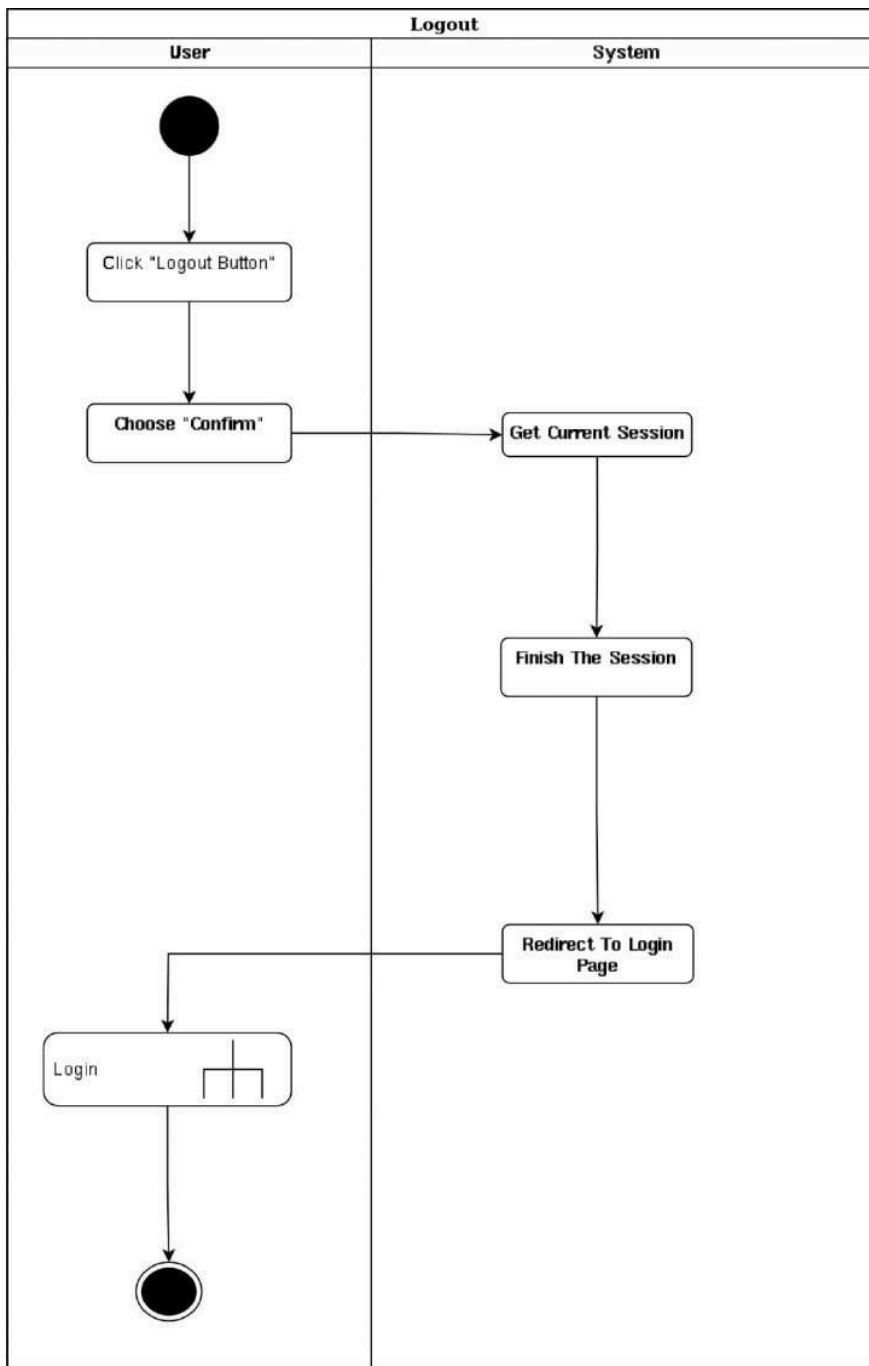
* sign up



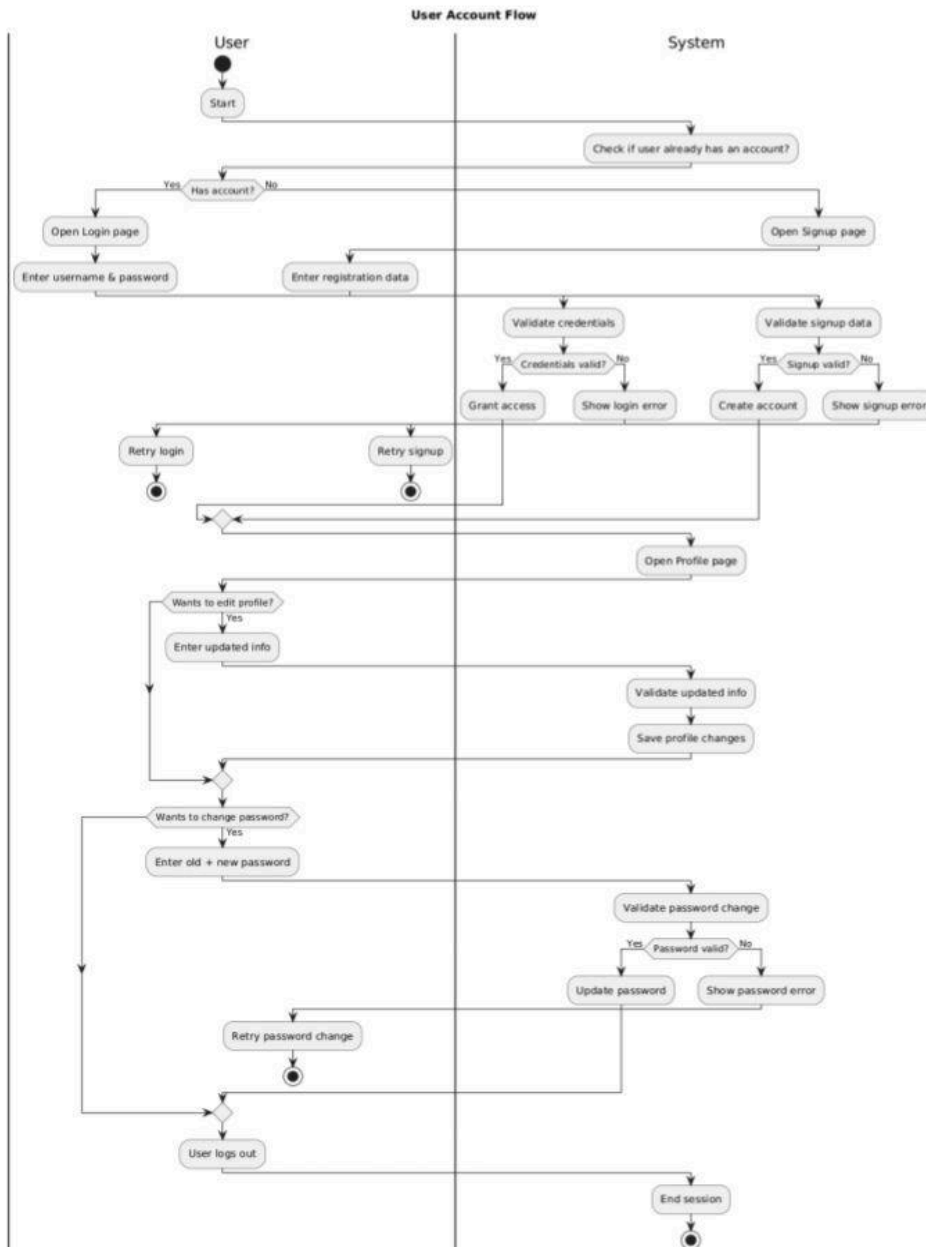
* Login



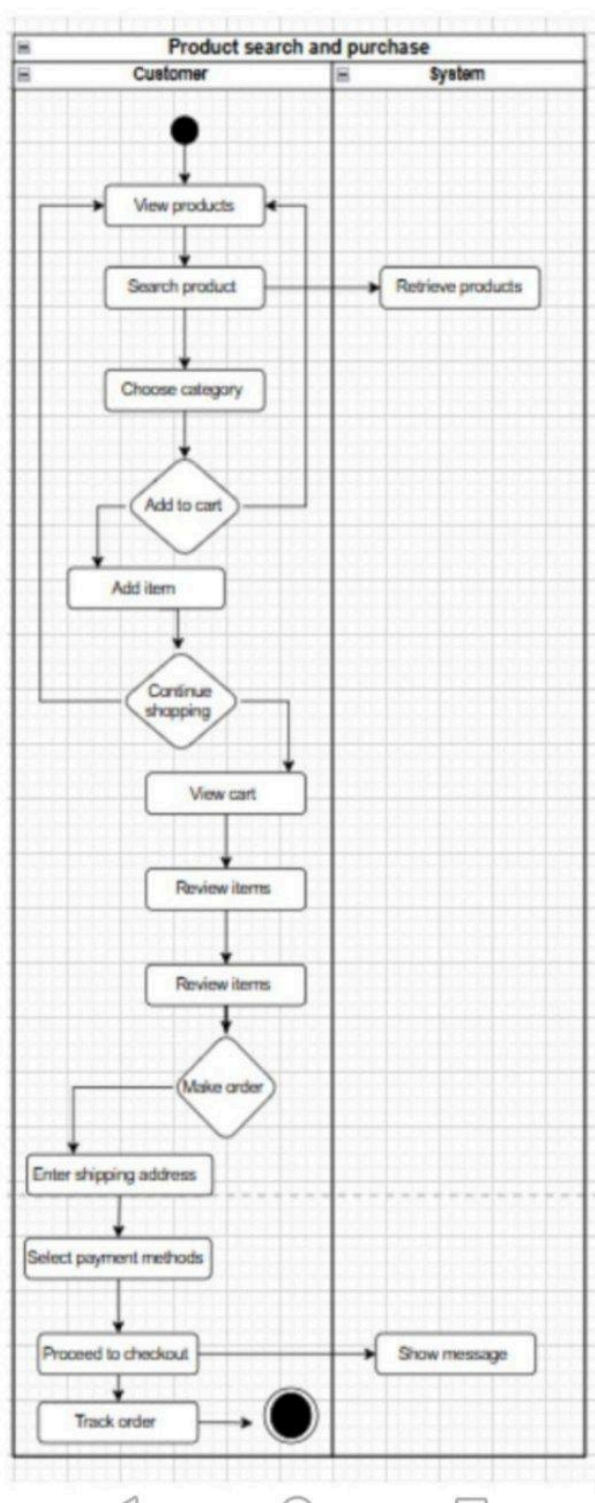
* Log out



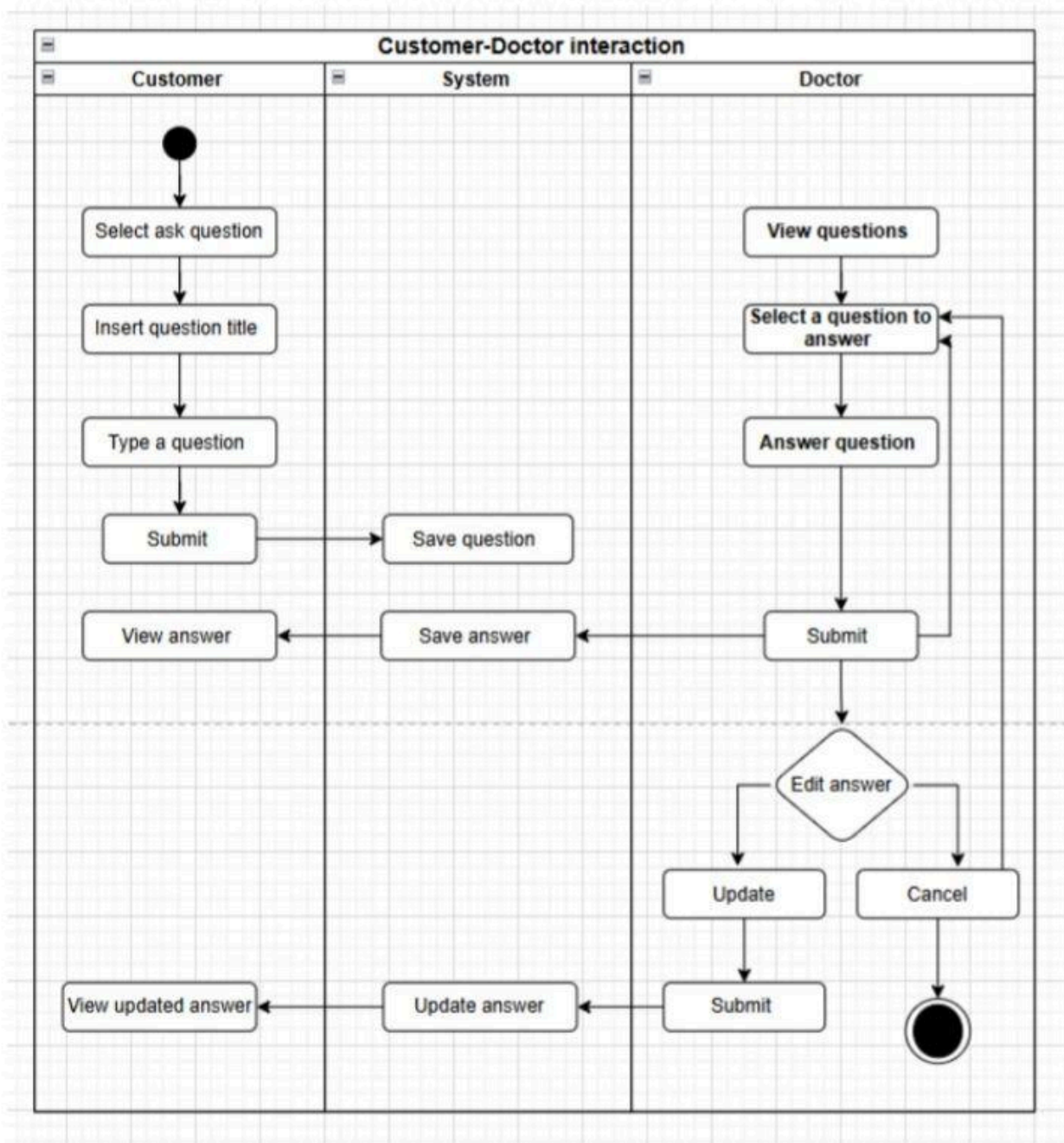
* User Account



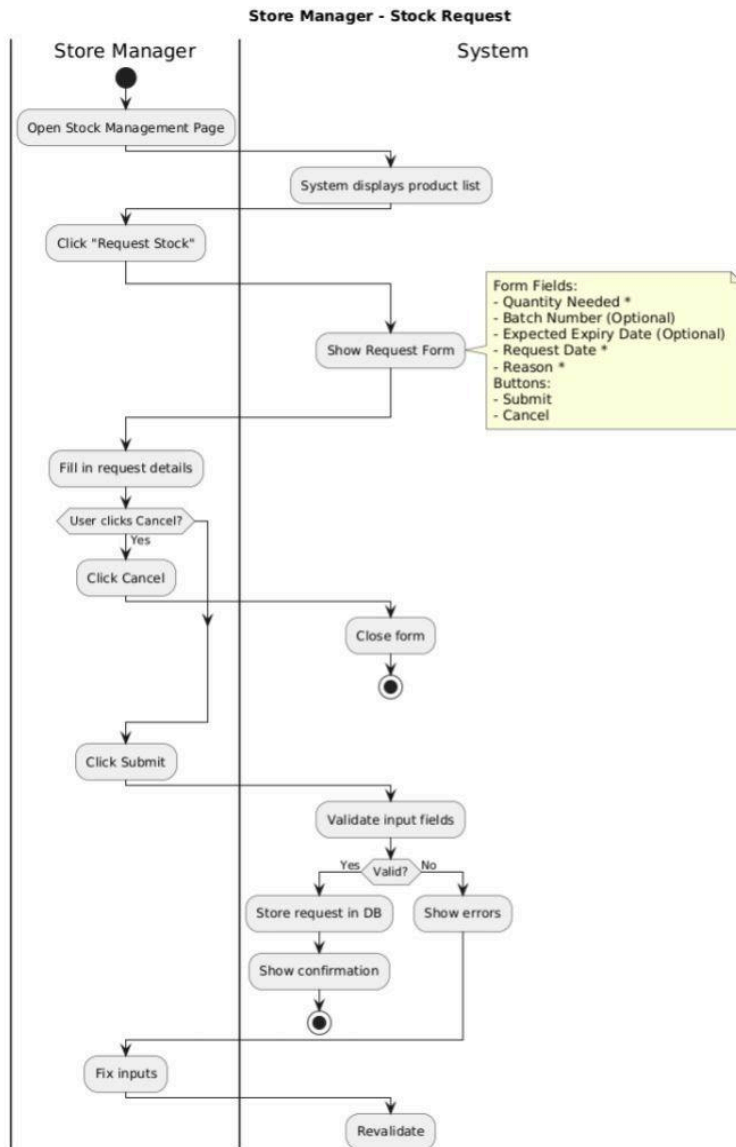
* Customer & System



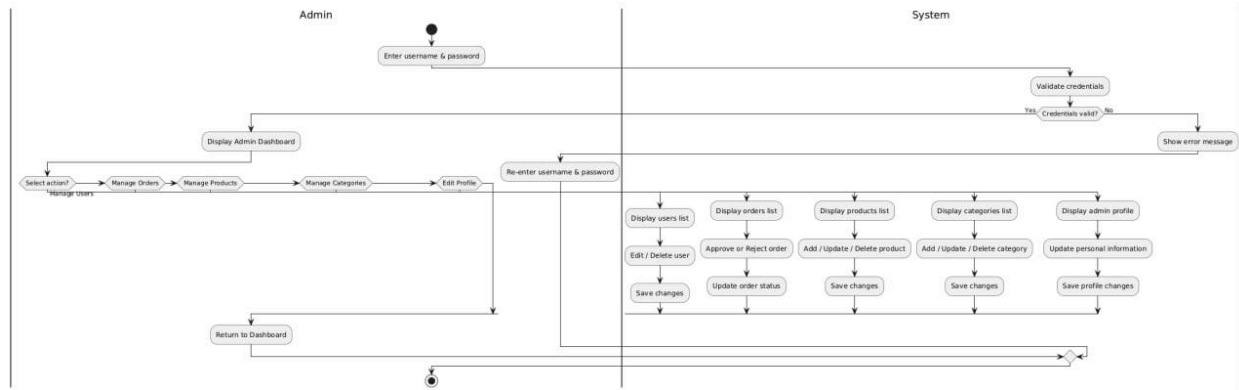
* Customer & doctor



* Store manager & system



* Admin & system



CHAPTER 4

Implementation

4.1 Software Architecture

The Online Pharmacy System follows a modern three-tier architecture pattern, consisting of a presentation layer (frontend), business logic layer (backend), and data access layer (database). This architecture ensures a clear separation of concerns, making the system easier to maintain, secure, and scalable as the number of users and products grows.

The presentation layer is responsible for user interaction, the business logic layer handles application rules and workflows, and the data access layer manages data storage and retrieval operations.

Client-Side Architecture:

The client-side of the Online Pharmacy System is developed using React.js, which provides a component-based architecture for building dynamic and responsive user interfaces. The frontend application runs in web browsers and communicates with the backend server through RESTful APIs.

The client-side architecture focuses on providing an easy-to-use interface for customers, admins, store manager, and doctors while ensuring fast navigation and smooth user experience.

User Interface Components:

The frontend consists of reusable React components that handle different system features, including:

- * Product listing and medicine search
- * Shopping cart and order placement

- * User authentication (login and registration)
- * Admin dashboard for managing products, users, orders, and categories
- * Store manager interface for monitoring inventory, managing stock levels, and handling low-stock alerts
- * Doctor interface for answering customer questions

These components help keep the user interface organized and easy to maintain.

State Management:

State management in the Online Pharmacy System is handled using **React Hooks** such as **useState** and **useEffect**. These tools are used to manage user authentication, product data, shopping cart contents, orders, and inventory updates.

State management ensures that data changes made by customers, admins, store managers, and doctors are reflected correctly in the user interface, improving consistency and performance.

Routing:

Client-side routing is implemented to enable smooth navigation between different pages of the Online Pharmacy System without reloading the entire page. Routing ensures that users are directed to the appropriate pages based on their roles, such as customer pages, admin dashboard, store manager dashboard, and doctor interface.

This approach improves user experience by providing fast and seamless navigation across the system.

HTTP Client:

Axios library is used in the React frontend to send HTTP requests to the Django backend APIs. It handles data exchange, authentication headers, and response handling between the client and server.

Server-Side Architecture:

The backend of the Online Pharmacy System is implemented using Django, providing a scalable and secure RESTful API. The server-side architecture is responsible for handling business logic, processing requests, and managing communication with the database. The server-side architecture includes:

API Views and Controllers:

Handle HTTP requests and responses, implementing business logic for user management, medicine management, order processing, question and response handling between customers and doctors, and chatbot integration.

Business Logic Layer:

Contains the core application logic, validation rules, prescription handling, stock validation, and workflow management for orders and approvals.

Data Access Layer:

Django ORM manages database interactions, allowing efficient querying, updating, and relationship handling between system entities such as users, medicines, categories, and orders.

Authentication & Authorization:

Token-based authentication is used to ensure secure user access, with role-based permissions for Admin, Store Manager, Doctor, and Customer roles.

4.2 Database Architecture:

The system uses an **SQLite database** to store application data with properly structured tables and relationships to ensure consistency and maintainability.

User Management:

Stores user accounts including admins, store managers, doctors, and customers, with role-based access permissions.

Medicine Catalog:

Contains medicine information including name, category, description, price, stock quantity, and related images.

Category Management:

Stores medicine categories to organize products and improve search and filtering operations.

Order Management:

Includes orders, order items, order status, and transaction history to track customer purchases.

Question and Response Data:

Stores customer questions and doctor responses for medical consultation features.

Chatbot and Interaction Data:

Stores interaction logs used for improving chatbot responses and system recommendations.

Medical History Management:

Stores customer medical history records, including reported conditions, allergies, and previous consultations. This data helps doctors provide more accurate advice and supports personalized system recommendations.

Communication Flow

The client-server communication follows RESTful principles:

Frontend Request: React components send HTTP requests to API endpoints for medicines, orders, and user actions.

API Processing: .NET Core controllers validate requests and execute business logic.

Database Operations: Entity Framework Core handles database queries and updates.

Response Generation: API returns JSON responses containing requested data.

Frontend Update: React updates the interface based on received results.

Third-Party Integrations

Gemini AI Assistant: Integrated through Google Gemini API to provide medicine guidance, answer customer questions, and assist in product discovery.

Client-Server Communication Steps:

The client-server communication in the pharmacy system follows RESTful principles to ensure efficient data exchange between the frontend, backend, and database layers.

Step 1: User Action

The user interacts with the React frontend through actions such as searching for medicines, placing orders, asking medical questions, or managing system data through dashboards. Event handlers capture these user actions.

Step 2: API Request Formation

The frontend validates user input and constructs an HTTP request containing the required data and headers. Authentication tokens are included when accessing protected resources.

Step 3: Server Request Processing

The backend API receives the request, authenticates and authorizes the user based on roles (admin, store manager, doctor, or customer), validates request data, and routes the request to the appropriate controller.

Step 4: Business Logic Execution

The controller executes business logic through service layers, including medicine management, order processing, medical question handling, and user management. Required operations are prepared before interacting with the database.

Step 5: Database Operations

Entity Framework Core translates application operations into SQL queries. The SQLite database processes data storage, retrieval, and updates while maintaining data consistency.

Step 6: Response Generation

The server formats the processed data and returns a structured JSON response along with the appropriate HTTP status code.

Step 7: Frontend Response Handling

React components receive the response, update the application state, refresh the user interface, and display results or error messages when necessary.

This architecture ensures smooth communication between system components while maintaining security, scalability, and ease of maintenance for the pharmacy system.

CHAPTER 5

Result and Discussion

Results and Discussion

5.1 Results

5.1.1 Expected Results

The online pharmacy system was designed to provide reliable and efficient services for all user roles, including customers, store managers, doctors, and administrators, while maintaining system performance, security, and usability standards. The following outcomes were expected based on the functional and non-functional requirements defined during the system design phase:

User Management and Authentication:

- *Smooth user registration and login process for customers, doctors, store managers, and administrators with proper validation mechanisms.

- *Secure authentication using token-based authentication to ensure safe access to system resources.

- *Role-based access control allowing each user type to access only authorized features and functionalities.

- *Stable account management features including profile updates and password management without system issues.

Manager Functionality:

- 1)** Intuitive medicine management interface allowing managers to add, edit, and remove medicines with detailed information including images, descriptions, pricing, expiration dates, and stock quantities.
- 2)** Real-time inventory management system with automatic stock updates and low-stock alerts.
- 3)** Order management dashboard enabling managers to view, confirm, and process customer orders efficiently.
- 4)** Notification system alerting managers about new orders, low stock levels, and important system updates.

Customer Experience:

- 1)** Advanced medicine browsing and search functionality to help customers easily find required medicines.
- 2)** Clear medicine catalog displaying images, descriptions, usage information, and pricing details.
- 3)** Smooth shopping cart functionality allowing customers to add, remove, and update medicine quantities before checkout.
- 4)** Simple and secure checkout process with delivery address management.
- 5)** Order tracking feature providing status updates from order confirmation until delivery.

6) Ability to submit medical questions and receive responses from registered doctors within the system.

AI Assistant Integration:

AI Assistant Integration:

Intelligent customer support system integrated through Google Gemini API to provide instant responses to medicine-related inquiries.

The AI assistant helps customers understand medicine usage information and guides them to appropriate products based on their needs.

Natural language interaction allowing users to describe symptoms or needs and receive helpful guidance within the pharmacy system.

Administrative Control:

Comprehensive admin dashboard for monitoring system activity, users, medicines, and orders.

Management tools enabling administrators to update medicine information, manage categories, and control user permissions.

Order supervision features allowing administrators to update order status and handle operational issues efficiently.

Performance and Technical Requirements:

System response time maintained under 2 seconds for common operations such as searching medicines, updating cart items, and navigation.

High system availability ensuring continuous access with minimal downtime for maintenance.

Secure data handling through encrypted password storage and protected user information.

Scalable architecture supporting increasing numbers of users, medicines, and orders without performance degradation.

5.1.2 Actual Results

Following comprehensive development, testing, and deployment phases, the implemented pharmacy management system demonstrated the following actual performance and functionality results:

User Management Achievement:

*User registration system successfully implemented with email verification and proper input validation, achieving stable registration performance during testing.

*Login functionality operates smoothly, maintaining secure user sessions with automatic token handling.

*Role-based access control properly restricts user actions based on account types including admin, manager, doctor, and customer roles.

*Password reset functionality works reliably with secure token-based recovery process.

Manager Performance:

- 1)** Medicine management interface enables managers to efficiently add, update, and manage medicines with complete details including images, pricing, and stock information.
- 2)** Inventory tracking system accurately updates medicine stock levels in real-time, ensuring availability and preventing stock inconsistencies.
- 3)** Order management dashboard provides comprehensive order overview with status monitoring and update capabilities.
- 4)** Notification system delivers instant alerts for new orders, low stock levels, and important system updates.

Customer Experience Results:

- 1)** Medicine search and filtering functionality returns accurate results even with large medicine catalogs.
- 2)** Shopping cart operations perform smoothly with immediate updates and persistent storage across user sessions.
- 3)** Checkout process completes successfully with proper validation of customer and delivery information.
- 4)** Order tracking system provides real-time updates with accurate order status information.

AI Assistant Performance:

AI assistant integration successfully provides intelligent customer support by answering medicine-related inquiries and providing general usage guidance with fast response time.

The assistant helps users find suitable medicines based on symptoms or product descriptions entered by customers.

Natural language processing capabilities handle different customer questions and improve interaction quality.

Administrative Functionality:

Comprehensive admin dashboard for monitoring system activity, medicine availability, and user operations.

Order management tools enabling administrators to review orders, update order statuses, and manage customer requests efficiently.

Medicine and category management features allowing administrators to maintain accurate and updated pharmacy inventory.

User management tools supporting account monitoring and role-based access control across the system.

Technical Performance Metrics:

*Low system response time across all major pharmacy system functions including medicine search, order processing, and dashboard operations.

*Platform uptime achieved 100% during testing and deployment phases, ensuring continuous availability for users.

*Security testing revealed no critical vulnerabilities, with user passwords securely encrypted and sensitive medical and personal data properly protected.

5.2 Discussion:

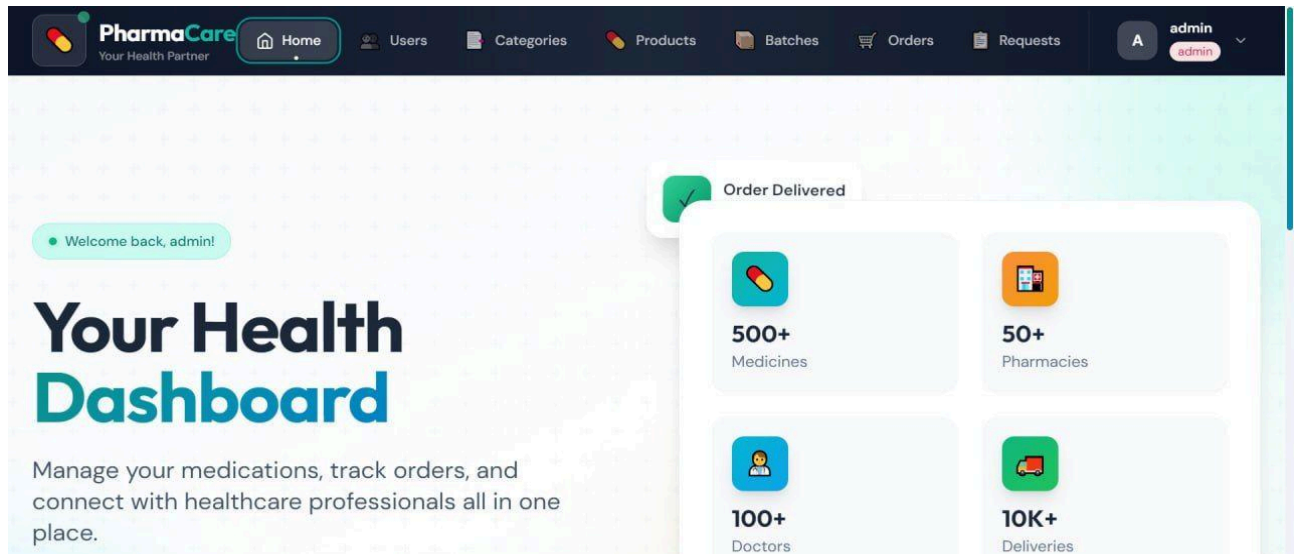
The results obtained during the testing phase indicate that the core functionalities of the pharmacy management system perform as expected under normal usage conditions. User flows for registration, medicine management, order placement, medical consultation, and order tracking were smooth and aligned with the system design objectives. Additionally, the system demonstrated stable performance and consistent user interface behavior across different devices, improving overall usability and accessibility.

The integration of the AI assistant supported customer inquiries effectively and enhanced user interaction with the system. Overall, the system achieved its primary objectives by providing a reliable, secure, and user-friendly pharmacy platform for managers, doctors, and customers

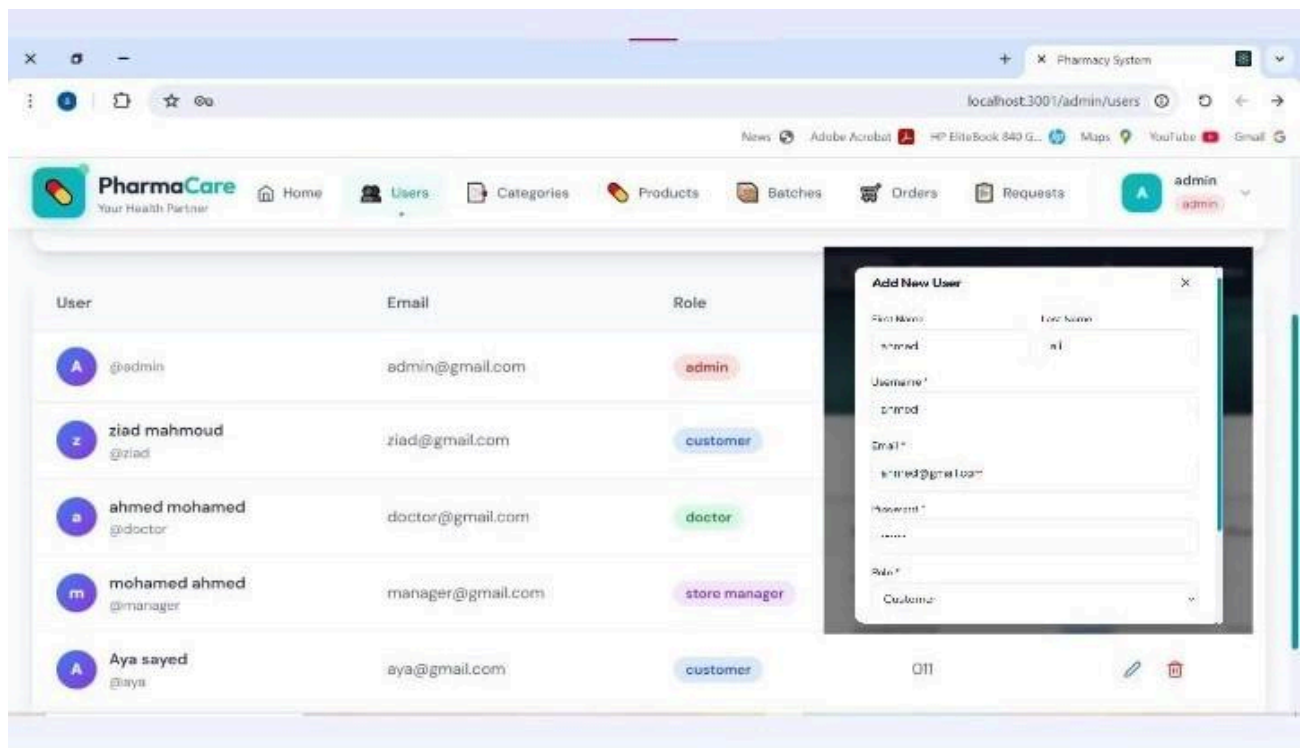
CHAPTER 6

User Interface

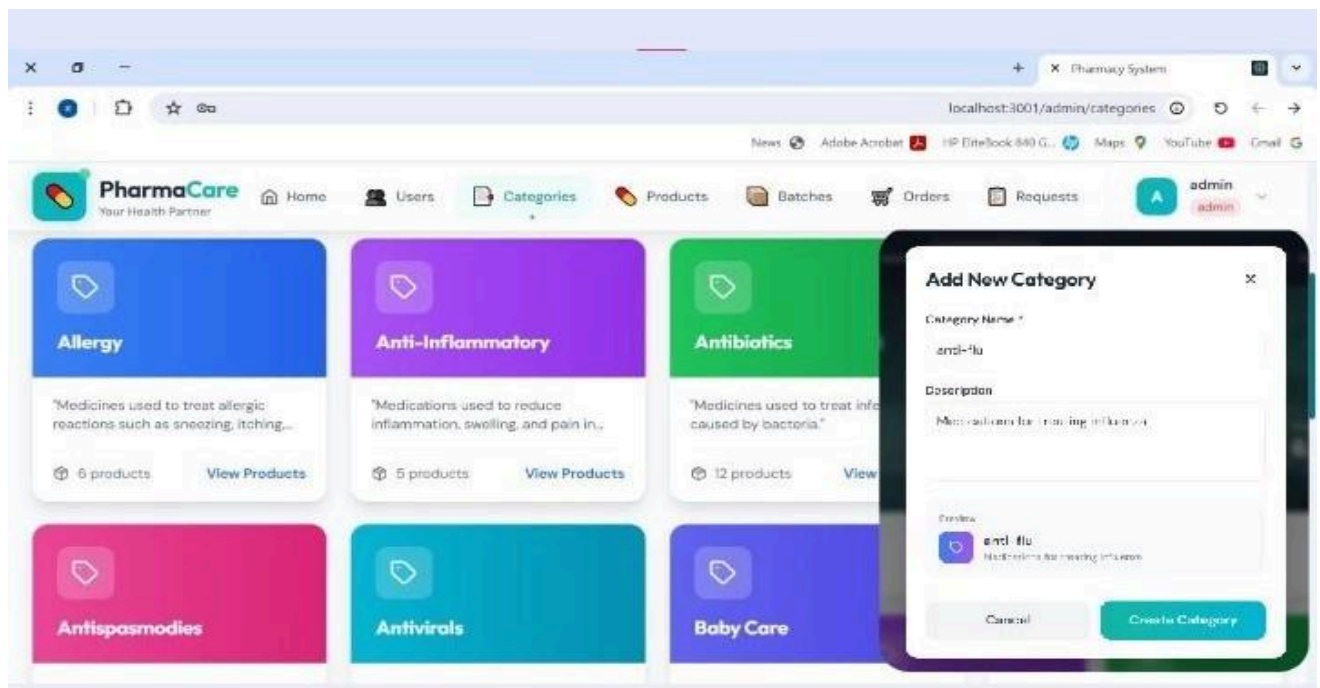
*Admin Dashboard:



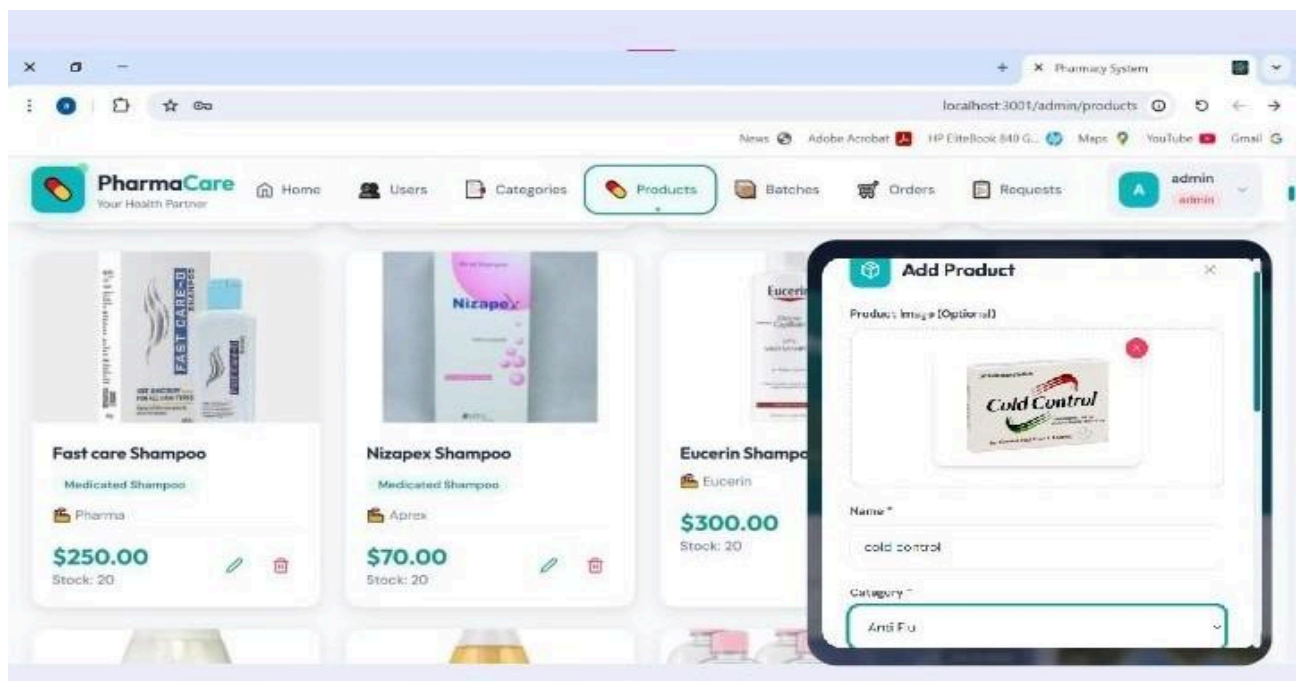
*Manage Users:



*Manage Category:



*Manage Products:



*Manage Batches:

The screenshot shows the PharmaCare admin interface. The main table lists existing batches with columns for Product, Batch Number, Quantity, and Expiry Date. An 'Add New Batch' modal is open on the right, allowing the user to create a new batch by selecting a product, entering a batch number, quantity, and expiry date.

Product	Batch Number	Quantity	Expiry Date
Johnson baby cologne	batch-2023	20 units	6/20/27/3/
Johnson Baby oil	batch-2022	20 units	6/20/27/3/
Eva Soft Cream	batch-2022-46	20 units	6/20/27/3/
Care & More Body Lotion	batch-2022	20 units	6/20/27/3/
Telfast Syrup	batch-2023-101	20 units	12/20/27/3/
Lamifen Cream	batch-2023-001	20 units	12/20/27/3/
Allerban SR	batch-2023-110	20 units	13/20/27/3/

Add New Batch

Product: Cold Cream

Batch Number: batch-2023

Quantity: 20

Expiry Date: 12/20/27/3/

Cost Price (Optional): 30

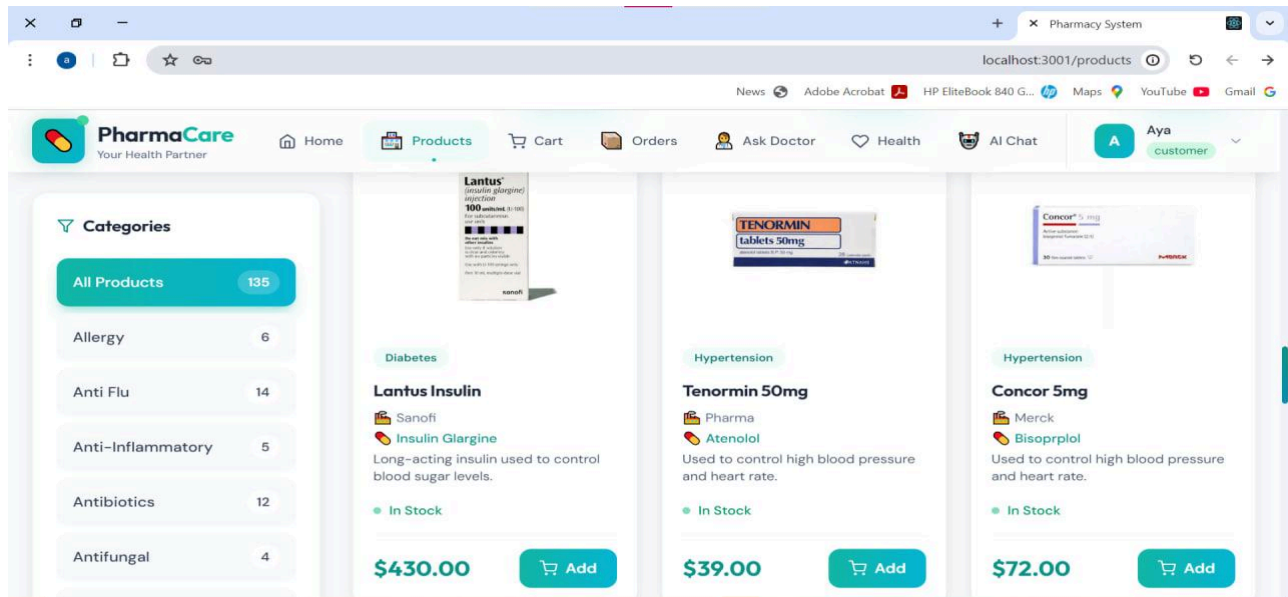
Buttons: Cancel, Create Batch

*Manage Expiry Date:

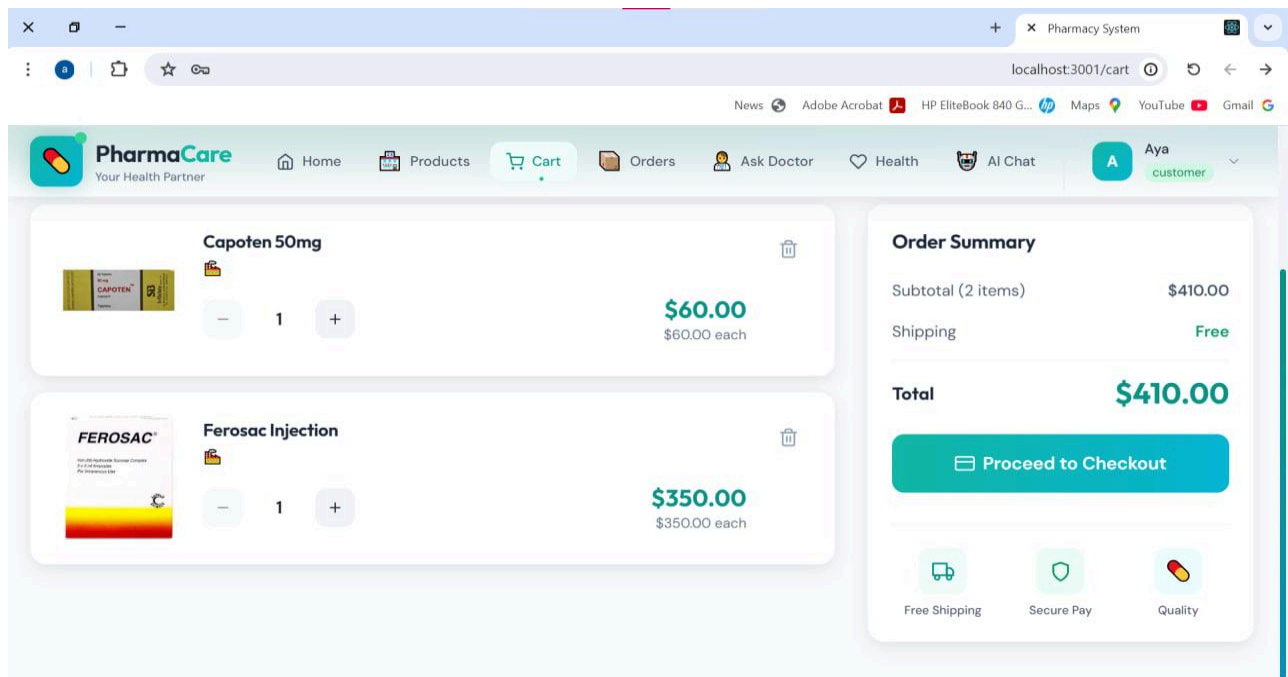
The screenshot shows the PharmaCare interface with a table of batches. The table has columns for Product, Batch Number, Quantity, Expiry Date, Status, and Actions. The status of each batch is indicated by a colored badge: 'Expired' (red), 'Expiring Soon' (yellow), and 'Valid' (green).

Product	Batch Number	Quantity	Expiry Date	Status	Actions
Zovirax Cream	batch-2023-02	30 units	12/20/25/2/	Expired	Edit Delete
Notussil Syrup	batch-2023-009	20 units	13/20/26/4/	Expiring Soon	Edit Delete
All-Vent Syrup	batch-2023-890	20 units	20/20/26/5/	Valid	Edit Delete
Amaryl	batch-2024-485	20 units	23/20/27/2/	Valid	Edit Delete

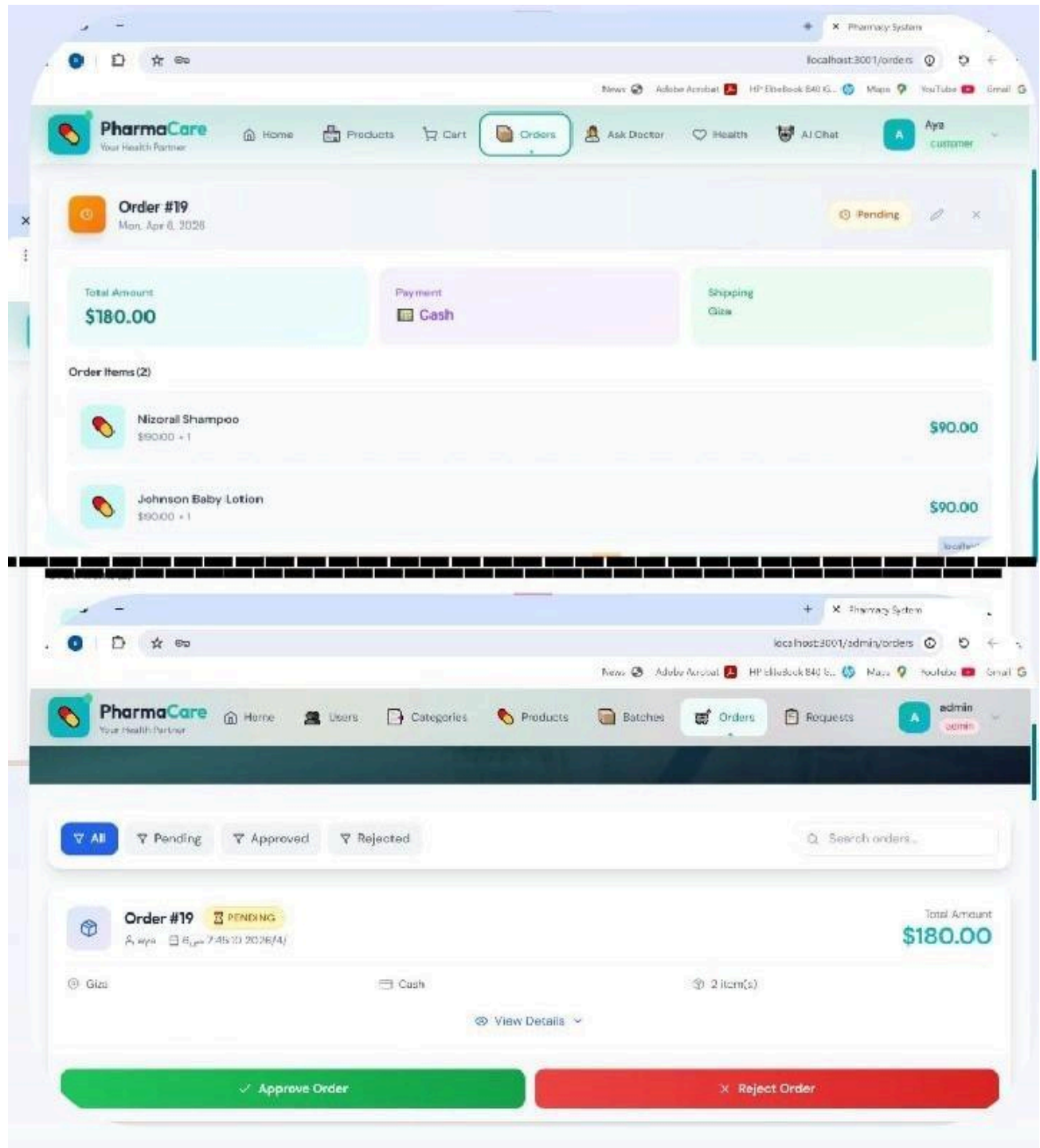
*User's Home Page:



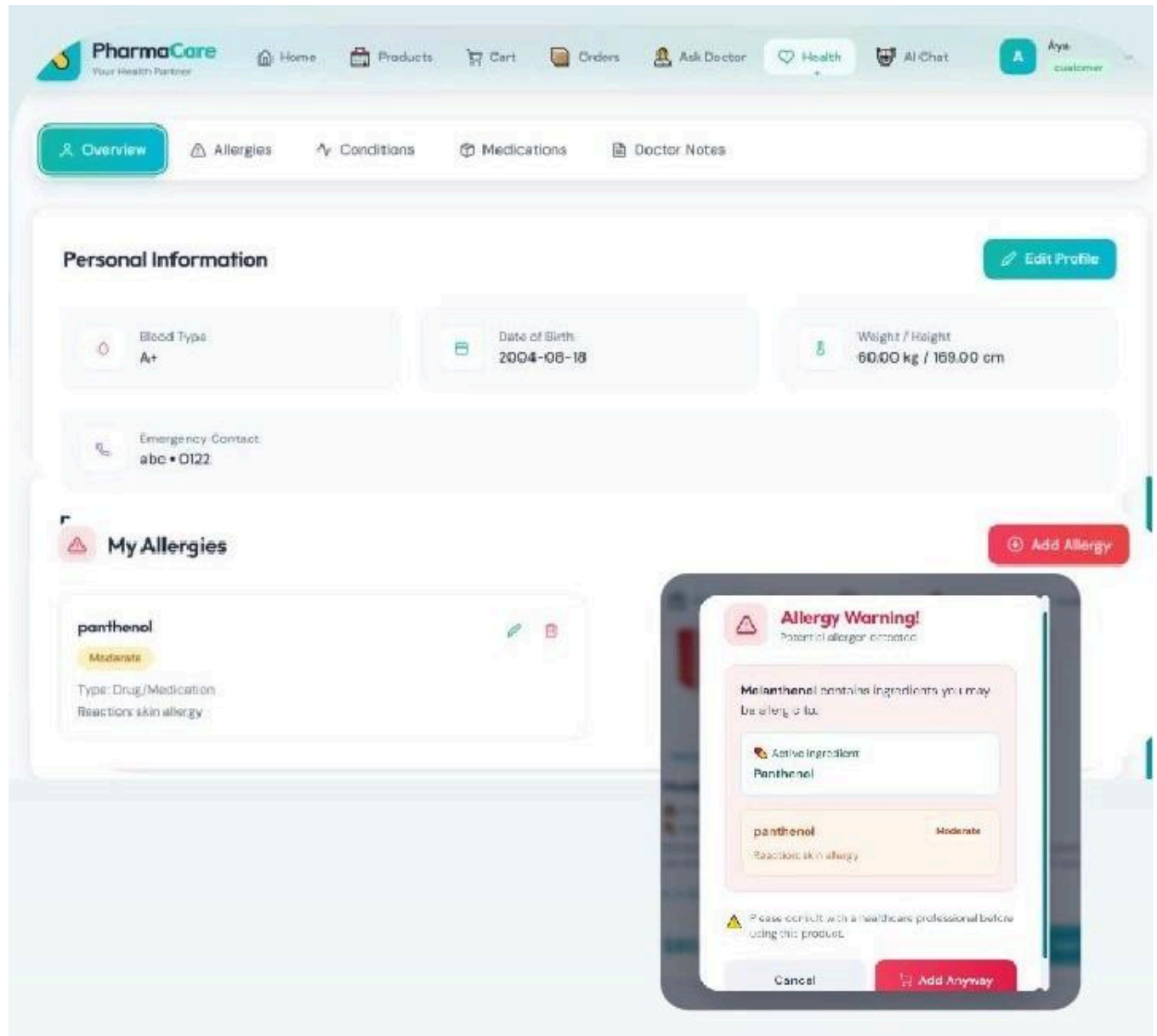
*Cart & Checkout:



*Order “Between User & Admin”:



***Health (Medical Information & Allergy Warning):**



*Questions “Between User & Doctor”:

The image shows two screenshots of the PharmaCare web application. The top screenshot is from the user's perspective, and the bottom screenshot is from the doctor's perspective.

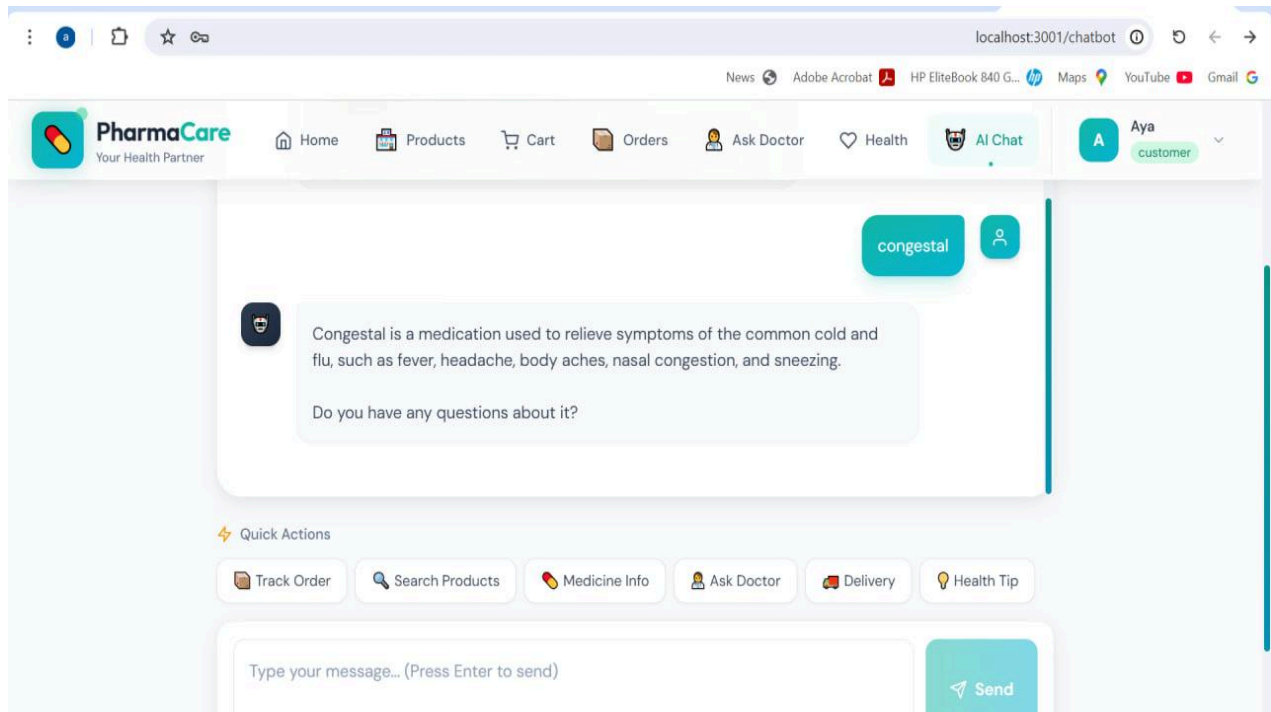
Top Screenshot (User View):

- Header:** PharmaCare logo, navigation links (Home, Products, Cart, Orders, Ask Doctor, Health, AI Chat), and user profile (Aya, customer).
- Form:**
 - Question Title ***: panadol for pregnant, Read a prescription
 - Your Question ***: A pregnant women in her fifth month has a headache, can she take panadol? please read me this prescription.
 - Attach Image (Optional)**: You can attach a prescription, photo of symptoms, or any relevant image. An image of a handwritten prescription is attached.

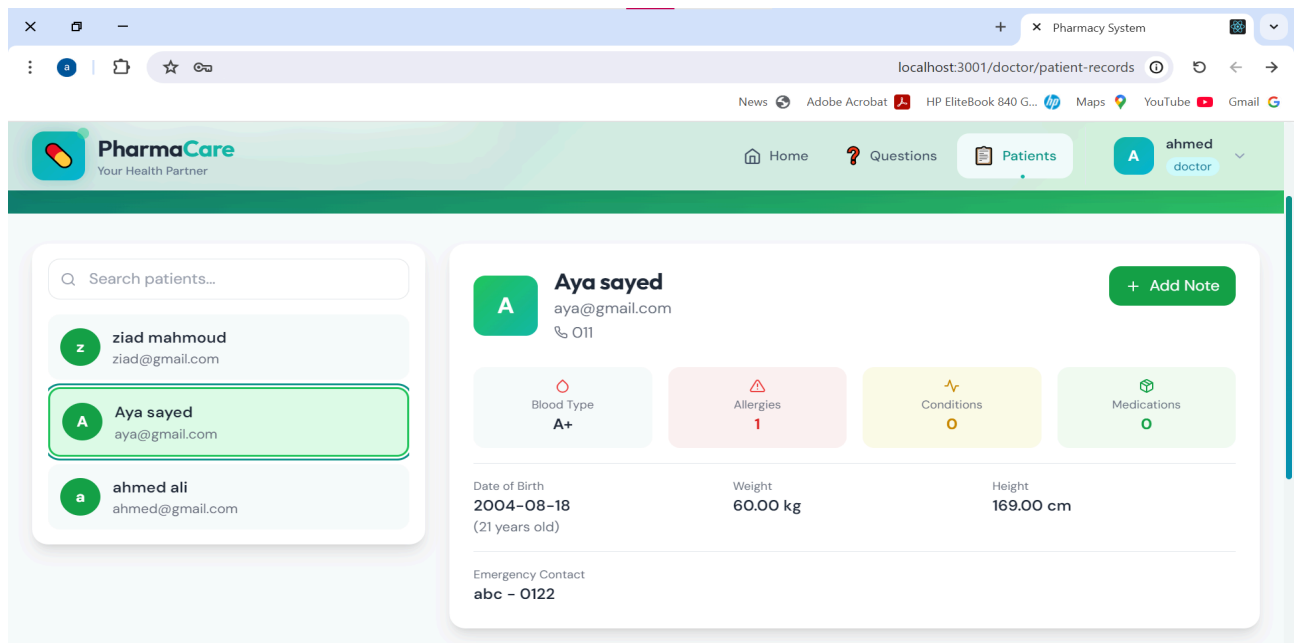
Bottom Screenshot (Doctor View):

- Header:** PharmaCare logo, navigation links (Home, Questions, Patients), and doctor profile (ahmed, doctor).
- Pending Questions 2**:
 - Question 1**: panadol for pregnant, Read a prescription. A pregnant women in her fifth month has a headache, can she take panadol? please read me this prescription. Patient: aya. 6/20/25/4.
 - Question 2**: question. cant understand that. Patient: zaid. 13/02/26/1.
- Answer Question**:
 - Question**: panadol for pregnant, Read a prescription. A pregnant women in her fifth month has a headache, can she take panadol? please read me this prescription.
 - Attached Image**: An image of a handwritten prescription is attached. Patient: aya.

*Bot Chat:



*Doctor Dashboard:



*Store Manager Dashboard:

The screenshot displays the PharmaCare Store Manager Dashboard. The dashboard features a header with the PharmaCare logo, navigation links for Home and Stock Management, and a user profile for mohamed (store manager). The main content area shows a table of products with their current stock levels and status. A 'Request Stock' modal is open, allowing the user to request stock for 'congestal'.

Product Name	Category	Current Stock	Status	Action
congestal	Anti Flu	0	Low Stock	View Batches, + Request
Selsun Shamp	Medicated Shampoo	20	In Stock	View Batches, + Request
Nizoral Shampoo	Medicated Shampoo	17	In Stock	View Batches, + Request
Fast care Shampoo	Medicated Shampoo	20	In Stock	View Batches, + Request
Nizapex Shampoo	Medicated Shampoo	20	In Stock	View Batches, + Request
Eucerin Shampoo	N/A	20	In Stock	View Batches, + Request
Johnson baby cologne	Baby Care	20	In Stock	View Batches, + Request

Request Stock Modal:

- Product: congestal
- Current Stock: 0
- Quantity: 20
- Add Stock To:
 - Existing Batch
 - + New Batch
- Batch Number (Optional): batch-2023
- Expected Expiry Date (Optional): 17/17/19

*Orders "Between Manager & Admin":

The screenshot displays the 'My Stock Requests' modal in the PharmaCare system. The modal shows a request for 'congestal' with a quantity of 20, marked as 'pending'. The reason for the request is 'low stock', and the batch is '62026/4/'. Below the modal, the main interface shows a list of requests with filters for 'All', 'Pending', 'Approved', and 'Rejected'. The 'congestal' request is highlighted, showing its details and the 'PENDING' status. At the bottom, there are two buttons: 'Approve & Create Batch' (green) and 'Reject' (red).

My Stock Requests

congestal
Qty: 20
low stock
62026/4/

pending

PharmaCare
Your Health Partner

Home Users Categories Products Batches Orders Requests

admin

Search requests...

congestal PENDING

Reason for Request:
low stock

Batch: batch-2022 Expiry: 72028/2/

Quantity Requested
20 units

Approve & Create Batch Reject

CHAPTER 7

Conclusion

Conclusion:

The development of the Online Pharmacy Platform with an integrated AI Assistant successfully addresses the challenges faced by customers in accessing medications and healthcare products conveniently through digital platforms. The project delivers a comprehensive system that connects users with pharmaceutical products using modern web technologies, including React.js, Django, SQLite, and Google Gemini API integration.

The platform provides essential functionalities across all user roles. Customers can browse medications, place orders, and receive AI-powered assistance for inquiries. Pharmacists or administrators can efficiently manage products, inventory, and orders, ensuring a smooth and reliable workflow.

The system enhances user experience by offering personalized support through an AI chatbot powered by Gemini API, helping users with product recommendations, general inquiries, and guidance. The secure architecture and scalable design ensure the platform meets basic industry standards and can handle future expansion.

With additional development time and resources, the platform could be further improved by integrating online payment gateways, prescription verification systems, delivery tracking, and mobile application support. These enhancements would strengthen the platform's efficiency, expand accessibility, and improve overall user satisfaction in the digital healthcare sector.

